

Feature-Optimized and Explainable Crop Recommendation Using an Evolutionary TabKANet Architecture

Abstract

Accurate crop recommendation is a critical component of precision agriculture, as inappropriate crop selection leads to reduced yield, inefficient resource utilization, and long-term soil degradation. Existing machine learning and deep learning approaches often suffer from redundant feature usage, weak modeling of nonlinear tabular relationships, and manually tuned hyperparameters, which limit their generalization capability. To address these challenges, this study proposes an intelligent crop recommendation framework that integrates Superb Fairy-wren Optimization Algorithm (SFOA) for feature selection, Draco Lizard Optimizer (DLO) for hyperparameter tuning, and an advanced Tabular Kolmogorov–Arnold Network with Transformer attention (TabKANet) for classification. The framework begins with robust data preprocessing, followed by evolutionary feature reduction using SFOA, nonlinear and dependency-aware learning through TabKANet, and adaptive hyperparameter optimization using DLO. Extensive experiments conducted on a publicly available agricultural dataset demonstrate that the proposed model achieves superior performance, attaining an accuracy of 98.37%, precision of 97.95%, recall of 97.88%, F1-score of 97.91%, specificity of 98.10%, and low error rates with RMSE of 1.20%, confirming its effectiveness, robustness, and suitability for real-world crop recommendation applications.

Keywords - Crop Recommendation, Precision Agriculture, Transformer Attention, Kolmogorov–Arnold Network, Optimization.

1. Introduction

Crop recommendation has become one of the decision-support systems in agriculture, affecting productivity, profitability, and sustainable land use. Wrong crop selection for a certain region or season may lead to yield losses and poor resource use. Most conventional statistical and regression techniques do not serve the purpose well because they don't model rainfall, temperature, soil nutrients, and irrigation patterns in terms of nonlinear and non-stationary relationships. Emerging out from these issues, recent studies found deep learning architectures beneficial, particularly Long Short-Term Memory (LSTM), which captures temporal dependencies. However, this is dependent on hyperparameter selection, which makes it interesting to incorporate bio-inspired optimization techniques like Gorilla Troops Optimization (GTO) to improve learning stability and prediction accuracy in crop recommendation and yield prediction tasks [1].

While crop recommendation accuracy has tended to a great extent to machine learning, the existing systems usually issue deterministic outputs ignoring the uncertainty in predictions. Outside of those systems, the noisy data from the sensor and climatic changes and incomplete data create that uncertainty observably, which decreases the quality of the given recommendation at the end of the day. As a solution, np based ensemble frameworks have been developed with entropy-driven quantification of uncertainty in probabilistic crop recommendations. Such confidence-backed systems would provide farmers with more trustworthy decision support by explicitly framing uncertainty in prediction, which further grants faith and robustness into sustainable agricultural planning. [2].

The progress in controlled-environment agriculture has truly extended the capacities of crop recommendation systems, especially in hydroponic farming. Hydroponics grants for the fine-tuning of nutrient solutions and environmental parameters; hence data-based optimization is ideal. Modern systems, aided by AI and IoT sensors, continuously track nutrient concentrations, temperature, humidity, and pH to suggest crops and adjustments to the parameters necessary for their growth. These intelligent AI IoT structures guarantee maximum efficiency in cultivation with minimal intervention

and advance sustainable production through intelligent real-time crop selection [3]. Crop recommendation platforms based on the cloud are efficient; however, the uninterrupted internet access required for these platforms and the centralized nature of computing give rise to latency, privacy concerns, and accessibility hurdles in remote agricultural regions. Based on these limitations, edge computing and Tiny Machine Learning (TinyML) have emerged as feasible alternatives. TinyML-based crop recommendation systems work offline and provide real-time decision support by deploying the optimized machine learning model such as Random Forest Classifier directly on low power microcontrollers. This localized processing paradigm offers fast, cost-effective, and privacy-preserving crop recommendations to advance precision agriculture in resource-constrained environments [4].

The increase in convoluted agricultural datasets has ushered in deep learning architectures to increase the recommendation accuracy concerning differing environment conditions. Having architectures such as Long Short-Term Memory (LSTM), Bidirectional LSTM (BiLSTM), and Transformer networks gives strength to learning the nonlinear interaction that exists within features and with long-range dependencies. These architectures enable the crop recommendation systems to adapt with dynamic climates and soil variants, thereby facilitating sustainable agricultural practices by matching crops with evolving environmental dynamics [5]. Specific to the agricultural conditions of different regions, particularly in developing nations with varying climate zones like India, soil nutrients, weather conditions, cultivation costs, and market indicators such as minimum support price would form parameters into which a crop recommendation system ought to be inserted. Supervised learning models such as Random Forest, Support Vector Machines (SVM), and time-aware modeling techniques have also been said to hold great promise for the provision of flexible crop suggestions across different states and varieties [[6]].

A major drawback of most contemporary crop recommendation methods is the disjoint treatment of soil characterization, where soil physical properties may be analyzed independently of soil chemical and nutrient properties. Soil texture, color, and structure affect crop suitability altogether with nutrient composition. To fill this gap, directed multimodal learning frameworks were introduced that fuse soil image-based classification and structured nutrient profiling. By simultaneously integrating the visual and chemical knowledge, these hybrid systems provide more tailored holistic representation of the soil for accurate, real-time crop recommendation in precision agriculture [7]. Machine learning-based crop recommendation also gained importance as a data-driven alternative to those experience-based farming practices. The algorithms such as Random Forest are able to capture the complex interplay between land state and crop requirements by analyzing soil parameters like pH, moisture, and nutrient status. These intelligent systems help farmers select crops better suited to their local soil characteristics, thus improving yield potential while minimizing resource wastage and supporting the environmentally sustainable choice of farming practices [8].

The incorporation of agricultural domain knowledge into neural networks-based crop recommendation models has been done to improve decision accuracy further. The focus of these data-based systems on essential soil parameters like nitrogen, phosphorus, potassium, and pH allows them to offer recommendations that are more tailored than generalized agronomic ones. AI and Internet of Things technologies enable continuous data acquisition and adaptive learning, which makes such systems all the more useful to smallholder farmers needing optimized and sustainable crop planning solutions [9]. With an ever-increasing abundance of historical agricultural and geographical data, it has become easier to create site-specific crop recommendation systems. While analyzing area-wise and season-wise crop production patterns, machine-learning-based decision-support frameworks can deliver relevant recommendations in a timely and localized manner. Using approaches such as the Synthetic Minority Oversampling Technique (SMOTE) for resolving data imbalance issues yields greater reliability to the model prediction. These data-driven approaches minimize reliance on manual skill and aid in making decisions on crop selection in consonance with regional productivity trends and seasonal variability [10].

This research primarily aims to address the growing requirement for intelligent decision support systems for agriculture that operate reliably with heterogeneous, high-dimensional, and interdependent data situations. Recent farming environments generate diversified tabular information about soil composition, climate variability, seasonal dynamics, and crop-specific requirements; however, it is a challenge to consolidate this information into accurate recommendations. Existing learning models mostly grapple with simultaneous capturing of nonlinear feature behavior, complex inter-feature dependencies, and optimal model configuration with minimum manual intervention. Hence, the further motivation of this research involves the design of a flexible unified learning framework-nearly optimization-driven-which can adapt to data complexity on its own, enhance predictive confidence, and provide consistent crop recommendations that can somehow be implemented in real-life precision agricultural practices.

Many machine learning and deep learning approaches have been explored for crop recommendation, but unfortunately, some real limitations are still present in most cases. Most of the existing systems treat feature selection, model learning, and hyperparameter tuning as independent stages and hence do not create the optimal interface for interaction between data representation and model optimization. Many deep models constructed for handling tabular agricultural data are based on fixed activation functions and do not have specific mechanisms in place to model feature-wise nonlinear transformations with global dependency learning. Attention-based and ensemble techniques can vastly improve accuracy at the expense of high computation cost or lack of adaptability in hyperparameter search. Thus, there is an apparent research gap for an integrated framework jointly performing adaptive feature selection, expressive nonlinear modeling, dependency-aware learning, and automated hyperparameter optimization in a cohesive yet computationally efficient manner.

- A comprehensive intelligent crop recommendation framework is developed by synergistically integrating evolutionary feature selection, adaptive hyperparameter optimization, and advanced tabular deep learning to address the limitations of conventional agricultural decision-support systems.
- A bio-inspired Superb Fairy-wren Optimization Algorithm is employed to identify the most influential soil, climatic, nutrient, and crop-related features, effectively reducing dimensionality while preserving critical agronomic information.
- A hybrid TabKANet architecture combining Kolmogorov–Arnold Networks and Transformer-based attention is designed to accurately model nonlinear feature transformations and complex interdependencies inherent in agricultural tabular data.
- An adaptive Draco Lizard Optimizer is utilized to automatically tune critical learning and architectural hyperparameters, eliminating manual intervention and improving model generalization.
- A diverse and publicly available crop recommendation dataset incorporating soil properties, climatic conditions, nutrient levels, crop characteristics, and agricultural practices is systematically analyzed to validate the scalability and robustness of the proposed approach.

The structure of the paper delineates the clear presentation of the intended crop recommendation study. In Section 1, the background of the research, motivation, and problem formulation has been mentioned. In Section 2, existing crop recommendation approaches have been discussed, and research gaps have been identified. In Section 3, methodology for data preprocessing, feature optimization, model design, and learning strategy has been presented regarding the proposed methodology. Section 4 presents the experimental setup along with the performance evaluation and analysis. Section 5 presents the conclusion of the study and directions for future research in developing intelligent scalable agricultural decision support systems.

2. Related Works

Traditionally, crop recommendation studies targeted the mapping of soil nutrients and climatic variables to classes of suitable crops through conventional supervised machine learning techniques. Various models have been studied extensively during this period, including Decision Tree, K-Nearest Neighbors (KNN), Random Forest, and Extreme Gradient Boosting (XGBoost); Random Forest appears to have gained a reputation as the more dependable model, due to its ensemble nature and robustness against feature variability. The approaches were robust enough to prove that combinations of different decision trees were able to capture nonlinear relationships between nitrogen, phosphorus, potassium, pH, temperature, rainfall, and humidity, thereby improving accuracy of recommendation. These systems, however, mostly relied on a static set of features and shallow learning mechanisms, which limited their flexibility under the complexities involved in agricultural conditions and consequently large-scale implementation [11]. To move beyond single classifier-dependent performance enhancements toward better predictive accuracy, ensemble-based learning frameworks were developed that integrate the output of more than one supervised model using voting and aggregation strategies. By providing ensembles with a set of Logistic Regression, Support Vector Machine, KNN, Decision Tree, Artificial Neural Network (ANN), and XGBoost models, the systems enhanced their robustness and generalization ability for diverse farming conditions. Such ensemble approaches have demonstrated that the aggregation of heterogeneous learners could alleviate the bias and variance of individual models and obtain more stable crop recommendations under region-specific agricultural scenarios [12].

As field data becomes more and more real-time, numerous studies have included IoT technologies to enable dynamic crop recommendations. These historical agricultural records and real-time sensor measurements allow the adaptive decision-making under conditions that might change in the soil and environment. Lightweight machine learning models such as KNN typically balance accuracy with low computational load, making it possible to seamlessly integrate web- or mobile-based interfaces for farmers and real-time advice support. To improve prediction accuracy and optimization efficiency further, overall bio-inspired optimization techniques were incorporated into the machine learning models. For crop recommendation, hybrids of Moth Flame Optimization with Machine Learning, or HMFO-ML, utilized Probabilistic Neural Networks (PNNs) while using the Extreme Learning Machine (ELM) for yield prediction, after tuning model parameters via optimization algorithms. These highlighted the power of evolutionary search strategies in improving convergence action and predictive outcomes over the manual tuning of models [14].

Besides classification-based recommendation systems, some studies have somehow incorporated regression-based models into the ensemble crop production forecasting and regional level planning. Long-term agricultural prediction tasks trialing hybrid architectures with KNN-Random-Forest-Ridge-Regression blending strategies have demonstrated superior robustness. Also, statistical validation techniques were used to ascertain ensemble model superiority, providing trustworthy crop recommendations for specific season-associated regional conditions [15]. As the great momentum of climate-adaptive agriculture developments came into play, deep learning models that capture temporal dependencies became the focal point. Although Temporal Convolutional Networks (TCN) were, by design, a more appropriate architecture to represent long-term climatic variations and their influences on crop suitability over recurrent architectures concerning stability and parallel computation, they did so quite effectively learning the nonlinear temporal patterns present in historical weather data in support of climate-resilient crop recommendation amidst ever-changing environmental conditions [16].

Knowledge-based and physics-informed approaches for enhancing model reliability and domain validity have been proposed. The Physics-Informed Neural Networks (PINNs) integrated agronomic constraints directly into the learning process, such as the optimal temperature ranges and soil pH thresholds. When combined with ensemble learners like Random Forest, Extra Trees, and XGBoost via stacking schemes, these models were able to generalize better while ensuring that the predictions adhered to known agricultural principles rather than data-driven patterns [17]. Detailed comparative studies evaluated different supervised machine learning algorithms over standardized crop

recommendation datasets. Ensemble classifiers, in particular, Random Forest and Gradient Boosting variants, have always shown better performance than linear and distance-based models by accounting for feature interactivity and noise. These outcomes stressed the need for ensemble learning in agricultural decision-support systems and further pointed out limitations that single-model approaches have [18].

They imbibed predictive decision-making by integrating weather forecasting modules as an extension of the frameworks. By marrying machine learning classifiers with Long Short-Term Memory (LSTM) networks for short-term climate prediction, they provide forward-looking crop recommendations according to forthcoming seasonal conditions. Such inclusion has improved planning accuracy and introduced practical tools, including crop calendars and interactive interfaces for the farmers [19]. As Hybrid ensemble architectures continued to evolve with the introduction of novel model combinations, such as Random Forest-XGBoost (RFXG), which aimed to exploit complementary strengths of tree-based learners. These systems showed that carefully designed ensemble fusion, alongside careful hyperparameter tuning and cross-validation, could significantly enhance classification reliability and robustness in multi-crop recommendation scenarios [20].

Attention-driven deep learning models only used for tabular data made a significant leap into interpretable architecture. TabNet does sequential attention while sparse feature masking, which removes the need for explicit feature selection, delivering interpretability by design. Coupled with associated explainability methods such as SHapley Additive exPlanations (SHAP), these models opened the door to a transparent venue for making sense of feature contributions at the levels of global and instance, addressing concerns of trust in agricultural AI systems [21]. From some studies that integrated remote sensing and deep learning for supporting climate-resilient agriculture at a broader spatial level, multispectral satellite imagery analysis was performed through Convolutional Neural Networks (CNN) for land suitability assessment while meteorological parameters were predicted by Recurrent Neural Network-Long Short-Term Memory (RNN-LSTM) models. Rule-based systems integrated spatial and temporal predictions for crop planning in specific regions under climate uncertainty [22].

Even though feature engineering and fusion approaches were used to boost classification accuracy, the resultant feature representations were combined into enriched stacks and voting ensembles constructed using PCA, Mutual Information, and Recursive Feature Elimination. These methods demonstrated enhanced stability and generalization on datasets of different scales and complexities [23]. Explainable AI (XAI) has gained prominence to make crop recommendation systems more transparent. The internals of the supervised models trained with Local Interpretable Model-Agnostic Explanations (LIME) provided human-understandable insights into model decisions, allowing farmers and agronomists to understand why certain crops were recommended for certain conditions [24].

Real-time decision support systems combine IoT hardware with machine learning analysis to offer recommendations for crops that are processed on the fly. Sensor-laden platforms captured soil nutrients and environmental conditions, while ensemble classifiers were used to process incoming data streams for instant recommendations through easy graphical interface, ostensibly in support of precision agriculture and sustainable farming practices [25]. Advanced deep learning architectures preceded by optimization techniques continued to improve performance and stability. CEGRU models modified the regular recurrent networks with a stricter feature recalibration mechanism and hyperparameter tuning was undertaken by metaheuristic algorithms such as Hunter–Prey Optimization, ultimately increasing accuracy, lowered error, and faster convergence [26].

Ensemble recurrent architectures optimized using bio-inspired algorithms further exemplified the merits of incorporating multiple temporal learners. Models combining Long Short-Term Memory, Bidirectional LSTM, and Gated Recurrent Unit networks, all optimized through RFO, achieved better performance on both crop recommendation and yield prediction tasks by reducing overfitting and

increasing generalization [27]. Recent research extended crop recommendation frameworks toward multi-task precision agriculture systems by jointly tackling crop recommendation and disease detection. Deep convolutional and graph-attention-based architectures processed soil parameters and plant imagery to provide comprehensive farming assistance and underscored the burgeoning trend toward integrated agricultural intelligence platforms [28].

Advanced sustainability-focused agriculture was again enhanced with the very comprehensive deep learning pipelines that integrated soil classification, weather prediction, and multi-crop recommendation. These systems, then hinged on fusing spatial-temporal learning, optimization algorithms, and cross-attention mechanisms, achieved reliable recommendations for many crops, benefiting from SHAP and LIME analyses related to interpretability [29]. Last but not least, explainable crop recommendation models build on transparency principles meant to underline the idea of human-centered AI. It enabled the XAI-CROP framework to bring into being the integrated feature selection, decision tree-based learning applications, and post hoc explanations that made recommendations accurate and interpretable. Such measures ensure the strengthening of trust and usability in actual agricultural decision-making [30]. Table 1 provides a consolidated comparison of earlier crop recommendation studies concerning the different methodologies adopted by the different authors along with their key strengths and the limitations inherent in them, thereby providing a clear contextual reference for the identification of methodologies' trends and research gaps with respect to the proposed work.

Table 1. Comparative summary of existing crop recommendation methodologies, advantages, and limitations

Authors	Methodology	Advantage	Limitation
Punitha and Geetha et al. [1]	GTO-optimized LSTM for crop recommendation and DBN for yield prediction	Improved decision accuracy through optimal parameter tuning	Higher computational complexity due to hybrid optimization
Alam et al. [2]	Ensemble ML with entropy-based uncertainty quantification	Provides confidence-aware crop recommendations	Prediction reliability depends on entropy threshold selection
Rahman et al. [3]	AIoT-based hydroponic crop recommendation using machine learning and sensor data	High accuracy crop recommendation	Full automation not completely implemented
Baishya et al. [4]	TinyML-based Random Forest Classifier optimized and deployed on ATmega328P for standalone crop recommendation	Enables real-time, offline crop recommendation with high accuracy on resource-constrained hardware	Slight reduction in accuracy due to aggressive model compression
Rao et al. [5]	Uses LSTM, BiLSTM, and Transformer models for crop recommendation based on soil and climate data	Improves crop selection accuracy using temporal and attention-based learning	High computational complexity
Steven Sam et al. [6]	Random Forest and SVM with Lag Variable-based time-series modelling	Handles temporal dependencies in crop recommendation	Model interpretability is limited due to ensemble learning
Pandey et al. [7]	Multimodal fusion of soil images and nutrient data using deep learning	Improves crop recommendation accuracy through holistic soil analysis	Requires integration of multiple learning modules

Verma et al. [8]	Random Forest-based crop recommendation using soil parameters	Provides accurate and adaptive crop recommendations	Limited discussion on real-time deployment challenges
Musanase et al. [9]	Neural network-based crop recommendation with rule-based fertilizer logic	Provides accurate, soil-specific crop guidance	Does not include dynamic environmental factors
Senapaty et al. [10]	SMOTE-based data balancing combined with 13 machine learning classifiers for area-wise and seasonal crop recommendation	Achieves high classification performance with SGDC outperforming other classifiers	Limited to historical and survey-based data without real-time intelligent automation
Vemulapalli et al. [11]	ML-based CRS using Random Forest, XGBoost, KNN, and Decision Tree	High predictive performance	Limited to traditional ML models
Anam Umera et al. [12]	Ensemble supervised ML with ANN and XGBoost	High prediction reliability	Increased computational complexity
Vidhya et al. [13]	Uses KNN-based machine learning integrated with IoT sensors and a web application to recommend crops based on soil and climatic conditions.	Provides user-friendly, low-cost, and accurate crop recommendations using real-time and historical data.	Depends on correct sensor functioning and reliable network connectivity for real-time data transmission.
Gopi et al. [14]	Hybrid HMFO with PNN and ELM	High prediction accuracy	Increased algorithmic complexity
Hasan et al. [15]	Ensemble ML using KNN, RF, and Ridge Regression (KRR)	Improves prediction accuracy and robustness	Does not include soil properties or economic factors
Eliazer et al. [16]	TCN-based deep learning for climate-adaptive crop recommendation	Captures complex temporal climate patterns	Limited model interpretability
Gunasekaran et al. [17]	Physics-informed stacking ensemble using PINN	Physically consistent and robust predictions	Limited incorporation of agronomic constraints
Prity et al. [18]	Ensemble-based crop recommendation using nine ML classifiers with Random Forest as the final model	Achieves very high prediction accuracy for crop recommendation	Does not explicitly model temporal variability in climatic conditions
Mahale et al. [19]	EM-based data imputation with Random Forest and LSTM integration	Combines weather forecasting with crop recommendation	Recommendations are limited to predefined crop labels
Afzal et al. [20]	Ensemble learning using RF and XGB for multi-crop recommendation	Improved decision-making through combined model strengths	Increased computational complexity due to ensemble architecture
Stella Mary Venkateswara et al. [21]	TabNet-based interpretable deep learning with SHAP and	Provides transparent and explainable fertilizer and crop recommendations	Requires deep learning expertise for deployment

	attention-driven feature selection		
Mohamed et al. [22]	Hybrid CNN–LSTM with rule-based crop recommendation	Integrates spatial and temporal intelligence for crop planning	High computational complexity
Ahmed et al. [23]	Feature fusion with stacking and voting ensemble classifiers	High accuracy with reduced overfitting	Increased computational complexity
Shastri et al. [24]	Gradient Boosting with XAI (LIME)	High interpretability of predictions	Model sensitivity to noisy input features
Nimmagadda et al. [25]	IoT-based real-time crop recommendation using Random Forest	Enables real-time, sensor-driven crop prediction	Limited explainability of model decisions
Venkataresh et al. [26]	CEGRU with HPO-based hyperparameter tuning	Improved feature learning using concurrent excitation	Increased architectural complexity
Gopi et al. [27]	Ensemble of LSTM, BiLSTM, and GRU optimized using Red Fox Optimization	Improved prediction accuracy through ensemble learning and metaheuristic tuning	Increased computational complexity due to ensemble and optimization process
Sravanthi et al. [28]	IoT-integrated MKGPCNN for crop recommendation and CGSAAN for disease prediction	High prediction accuracy with integrated fertilizer recommendation	System complexity increases due to multi-stage processing
Mohan M et al. [29]	DTBRCNN + SPLSTMGNN with POA and Cross-Attention fusion	High accuracy with low latency for real-time multi-crop recommendation	Increased architectural complexity due to multi-module integration
Shams et al. [30]	XAI-based decision tree with LIME explanations	Improved transparency in crop recommendation	Relies on rule-based interpretability structure

Crop recommendation based on agricultural tabular data is thus fundamentally altered by nonlinear relation among features and the inter-feature interactions dynamically. Soil nutrients, climate, indicators of seasonality, and parameters of the crop seldom determine suitability in isolation; they interact with one another, exhibiting dynamic behavior across regions and time. Traditional tabular models and even advanced neural architectures often fail to model these interactions appropriately, resulting in biasing the learning process, causing difficulties in generalizing the model, or making the model unstable when deployed in different agricultural contexts. This poses as a major roadblock in developing reliable, scalable, and highly accurate crop recommendation systems which are sensitive to real-world variability.

The model performances exhibit a high sensitivity to configurations of hyperparameters that sometimes are set according to manual heuristics or exhaustive search strategies. Such techniques are inefficient, computationally expensive, and in practice inapplicable for complex hybrid architectures. On the other hand, poorly tuned hyperparameters may negate the advantages of highly powerful learning models, leading either to slow convergence or to less-than-optimum accuracy. Thus, the central question investigated in this research is why there is no end-to-end intelligent framework that can, at the same time, optimize feature relevance, learning expressiveness, dependency modeling, and hyperparameter selection to deliver crop recommendations that are robust and consistently high-performing in support of decision-making in precision agriculture.

3. Overall Framework Architecture

The intelligent crop recommendation system under consideration works as an integrated optimization-driven deep learning framework for the structured processing of agricultural data to make highly accurate crop predictions. The architecture is designed as a sequence of processes, wherein raw agricultural data are cleaned, optimized, and learned on, thus capturing the intricate relationship between various soil, environmental, and climatic factors. The entire workflow from beginning to end of the proposed system is illustrated in Figure 1, which shows the interaction between the preprocessing, feature optimization, classification, and performance evaluation modules.

At the beginning, the crop prediction dataset is fed as input to the system, which is now capable of processing the data in many ways to enhance the quality and consistency of preprocessing. This stage contends with missing values, transforms categorical attributes into numerical ones, and applies scaling and normalization methods to ensure uniform distribution of data. These will mitigate bias arising from dissimilar magnitudes of features and prepare the dataset for effective learning in the coming stages. After preprocessing, by means of the Superb Fairy-wren Optimization Algorithm (SFOA), the cleaned dataset, in turn, enters the feature selection module. It deals with the selection of the most informative soil and environmental attributes while discarding redundant and less-influential features. SFOA reduces the dimensional complexity of the learning model and improves its discriminative capacity by optimizing the feature subset, which in turn leads to improved efficiency in terms of both prediction and computation.

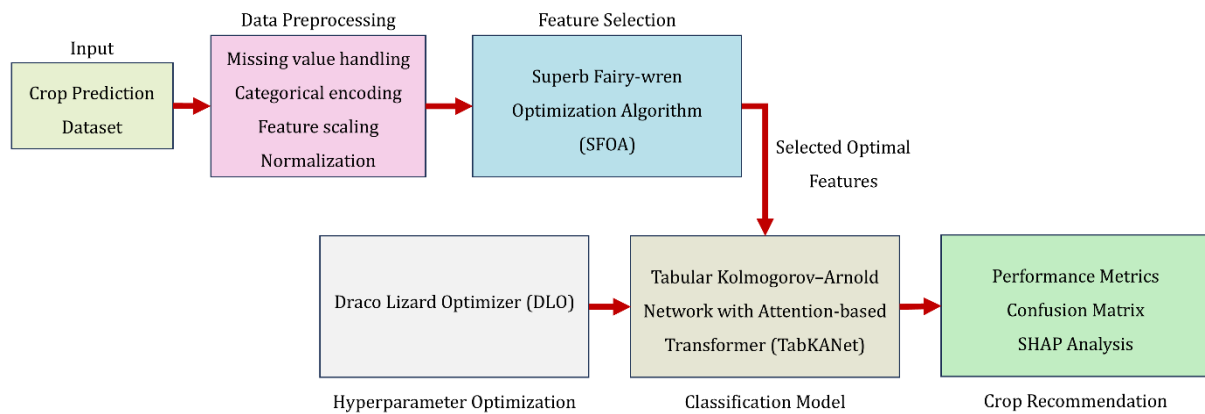


Figure 1. Overall architecture of the proposed SFOA–DLO–TabKANet-based intelligent crop recommendation framework.

The selected optimal feature set is piped to the Tabular Kolmogorov–Arnold Network with an attention-based Transformer architecture (TabKANet) for crop classification purposes. TabKANet exploits the vasculature of nonlinear relationship modeling and intricate feature interactions in such common agricultural tabular data. Besides this, to add to the learning and generalization competencies of the model, Draco Lizard Optimizer (DLO) has been put in place in the framework to adaptively optimize fundamental hyperparameters of the TabKANet architecture, which are responsible for major aspects like learning rate, network depth, and attention dimensions. The optimized TabKANet model further ends with crop recommendation output, which is tested through several performance assessment techniques. The standard classification metrics, confusion matrix analysis, and SHAP-based interpretability analysis systematically judge predictive accuracy and can trace a given feature contribution. This could only happen in so complicated a system as an evolved optimization algorithm with advanced deep learning that would allow the proposed framework to produce a scalable intelligent decision support for precision agriculture.

3.1 Data Preprocessing

Data preprocessing represents an exceedingly important phase in the crop recommendation framework proposed because agricultural datasets are often inflicted with various inconsistencies, heterogeneous types of features, and different ranges of numbers. If proper preprocessing is not performed, these problems become a great hindrance toward learning stability and predictive accuracy. In this study, an elaborate preprocessing pipeline is designed to improve data quality, achieve complete feature uniformity, and pave the way for efficient optimization and a deep learning-based model.

3.1.1 Handling Missing and Noisy Data

Often, real agricultural datasets contain missing entries due to reasons like sensor malfunctioning, incomplete soil testing, or human errors imposed during manual data collection. In order to mitigate the adverse effect of incomplete information, missing values for numerical attributes are estimated using statistical imputation methods. Here, mean imputation is used when the distribution of the data appears to be approximately symmetric, while median imputation is preferred for skewed distributions in order to make it less prone to outlier effects.

For a numerical feature, x , with missing values, it estimates its imputed value as:

$$x_i = \begin{cases} \frac{1}{N} \sum_{j=1}^N x_j, & \text{mean imputation} \\ \text{median}(x), & \text{median imputation} \end{cases} \quad \text{---} \quad 1$$

where N is the number of non-missing observations.

In addition to missing values, noisy data points caused by extreme jumps or misleading measures are made good by identifying the outlier observations using the interquartile range (IQR) analysis. Values that lie beyond the accepted limits are either capped or replaced with statistically representative values. Such cleansing of the noise avoids situations where abnormal data points dominate the learning process and hinder model training.

3.1.2 Categorical Encoding

Agricultural datasets generally have categorical attributes like soil type, irrigation mode, or region that directly cannot be processed by neural networks. Therefore, categorical variables are transformed into numerical representations via various encoding techniques. In this framework, label encoding is carried out on the ordinal categorical features, in which each category is assigned a unique integer value. One-hot encoding is used to ensure independence in the case of nominal attributes that do not possess any intrinsic order.

According to label encoding, a categorical feature $C = \{c_1, c_2, \dots, c_k\}$ is mapped as:

$$c_i \rightarrow i, i \in \{0, 1, \dots, k-1\} \quad \text{---} \quad 2$$

On the other hand, in the one-hot encoding, each category is represented in a binary vector:

$$c_i \rightarrow [0, \dots, 1, \dots, 0] \quad \text{---} \quad 3$$

Encoding is necessary for deep learning because it provides a mathematical representation for computational purposes, avoids incidental ordering bias for nominal features, and aids them in mixing well with the gradient optimization techniques that are used for training.

3.1.3 Feature Scaling and Normalization

Nutrient concentrations, rainfall and temperature are some of the agricultural characteristics that are very diverse in numerical scales. If left unscaled, therefore, those features on a larger magnitude would have a greater role in the learning of a model. For that reason, feature scaling and normalization were implemented to ensure that all attributes offer equal contribution.

Min-Max Normalization, is the rescaling of features, and by Min-Max normalization makes the range of values derived into a fixed interval [0,1]

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \quad \text{---} \quad 4$$

Alternatively, in the case of the features following near-Gaussian distributions, standardization is performed by which x is transformed to be of zero mean and unit variance:

$$x' = \frac{x - \mu}{\sigma} \quad \text{---} \quad 5$$

where μ and σ refer to mean and standard deviation of the relevant feature respectively. These transformations will enhance numerical stability, accelerate convergence during training and against gradient imbalance present in deep neural architectures such as TabKANet.

Algorithm 1: Data Preprocessing Procedure

Input: Raw crop prediction dataset D

Output: Preprocessed dataset D'

```

For each feature  $f$  in the dataset  $D$  do
  If feature  $f$  contains missing values then
    If the distribution of  $f$  is symmetric then
      Replace missing values with the mean of  $f$ 
    Else
      Replace missing values with the median of  $f$ 
    End If
  End If
  Identify outliers in feature  $f$  using the interquartile range (IQR) method
  Replace extreme values with the corresponding lower or upper boundary limits
End For
For each categorical feature  $c$  in dataset  $D$  do
  If feature  $c$  represents ordinal information then
    Apply label encoding
  Else
    Apply one-hot encoding
  End If
End For
For each numerical feature  $x$  in dataset  $D$  do
  If normalization is required then
    Apply Min-Max normalization
  Else
    Apply standardization
  End If
End For
Return the preprocessed dataset  $D'$ 

```

3.2 Feature Selection Using Superb Fairy-wren Optimization Algorithm (SFOA)

Feature selection plays a crucial role in intelligent crop recommendation systems, as agricultural datasets often contain correlated, redundant, and weakly informative attributes. The presence of such features increases computational burden and may degrade predictive performance by introducing noise into the learning process. To address this challenge, an evolutionary feature selection strategy based on the Superb Fairy-wren Optimization Algorithm (SFOA) is employed to identify a compact and informative subset of soil and environmental attributes.

3.2.1 Biological Inspiration and Optimization Principle

The small bird species named Super Fairy-wren is famous for its adaptive foraging behavior, exploration of territoriality, and collaborative decision-making to obtain food resources. These behaviors inspired SFOA to model candidate solutions in the form of individual birds which dynamically explore the search space while exploiting promising regions based on the information shared. This balance between wide exploration and focused exploitation offers a robust phenotype of SFOA as it navigates high-dimensional optimization problems like feature-subset selection.

In the proposed framework, each SFOA individual represents a potential feature subset encoded as a binary vector:

$$\mathbf{S} = [s_1, s_2, \dots, s_d], s_i \in \{0,1\} \quad \text{---} \quad 6$$

Where d denotes the total number of features. A value of $s_i = 1$ indicates that the i^{th} feature is selected, while $s_i = 0$ denotes exclusion. The binary encoding allows one direct control in the pursuit of dimensionality reduction while being efficient in evaluating candidate subsets.

3.2.2 Fitness Function Formulation

A multi-objective fitness function is designed to steer the search process towards optimal feature subsets whereby classification performance is maximized and selected features are minimally represented. The fitness value F for a solution is computed as follows:

$$F = \alpha \cdot \text{Acc}(\mathbf{S}) + (1 - \alpha) \cdot \left(1 - \frac{|\mathbf{S}|}{d}\right) \quad \text{---} \quad 7$$

where $\text{Acc}(\mathbf{S})$ denotes the classification accuracy obtained using the selected features; $|\mathbf{S}|$ is the number of selected features; d is the total number of original features; and $\alpha \in [0,1]$ is a weight factor controlling the trade-off between the maximization of accuracy and the reduction of features. The formulation favors selecting compact feature subsets without losing predictive capability.

3.2.3 Exploration and Exploitation Mechanisms

SFOA alternatively performs exploration and exploitation phases to maintain diversity within the population while converging toward high-quality solutions. Individuals update their positions randomly in relation to their neighboring solutions during the exploration phase, thus ensuring extensive coverage of the search space. This behavior is meant to avert the threat of premature convergence and ensure that multiple diverse combinations of features are assessed.

During the exploitation phase, individuals update their positions by moving toward the best-performing solutions identified during the search so far. This step of intensification accelerates convergence by further refining those feature subsets that exhibit better fitness. This adaptive swapping between the two phases enables SFOA to effectively balance global search and local refinement.

3.2.4 Feature Subset Selection and Termination Criteria

The optimization process continues to iterate and then stops when a predefined stop condition is reached. Termination occurs either when the maximum number of iterations is reached or when improvement in fitness is less than a specified threshold over successive iterations. The feature subset associated with the highest fitness score is selected as the optimal set and forwarded to the classification stage.

This evolutionary selection is a drastic reduction in dimensionality and removal of any redundant attributes, reducing to only the most salient soil and environmental features, thereby improving model efficiency and generalization.

Algorithm 2: SFOA-Based Feature Selection

Input: Preprocessed dataset D with d features

Output: Optimal feature subset S^*

Initialize a population of candidate solutions using binary encoding
Evaluate the fitness of each solution using the defined fitness function
Identify the best-performing solution in the population
While stopping criterion is not satisfied do
 For each individual solution in the population do
 Perform exploration by interacting with randomly selected solutions
 Perform exploitation by moving toward the current best solution
 Update the binary feature selection vector
 End For
 Recalculate fitness values for all updated solutions
 Update the global best solution if improvement is observed
End While
Return the feature subset corresponding to the best solution S^*

3.3 Crop Classification Using TabKANet

Accurate crop classification based on agricultural tabular data requires a learning model that can capture nonlinear transformations of the features and complex interdependencies among soil, climate, and seasonal variables. Traditional classifiers and classic neural networks very often fail to model such interdependencies due to limited representational flexibility. In this study, the Tabular Kolmogorov-Arnold Network with attention-based Transformer architecture is introduced to confront such challenges, combining functional approximation theory with self-attention mechanisms. The complete workflow of the TabKANet-based classification procedure is depicted in Figure 2, with considerations for its internal architectural components, which are elaborated upon in Figure 3.

3.3.1 Kolmogorov–Arnold Network (KAN) Component

The Kolmogorov-Arnold Network (KAN) is based on the Kolmogorov-Arnold representation theorem, which states that every multivariate continuous function can be expressed as a finite sum of univariate nonlinear functions and linear combinations. In this light, KAN captures nonlinearity by expressing high-dimensional mappings as other learnable functional transformations that are applied over the edges of the network instead of using fixed activation functions.

Mathematically, a multivariate function $f(\mathbf{x})$ with input $\mathbf{x} = [x_1, x_2, \dots, x_n]$ is represented as:

$$f(x) = \sum_{q=1}^Q \phi_q \left(\sum_{p=1}^n \psi_{qp}(x_p) \right) \quad \text{---} \quad 8$$

where $\psi_{qp}(\cdot)$ and $\phi_q(\cdot)$ are learnable univariate nonlinear functions. KAN can assign a different nonlinear mapping to each connection, thereby greatly increasing expressiveness, whereas conventional multilayer perceptrons would apply the same activation function across neurons. The capacity of having such functions is particularly suitable for agricultural tabular data, where relationships among features such as nutrient content, temperature, rainfall, and crop suitability are inherently nonlinear and heterogeneous. Thus, by learning feature-wise functional transformations, KAN is able to capture subtle agronomic patterns often missed by standard neural architecture.

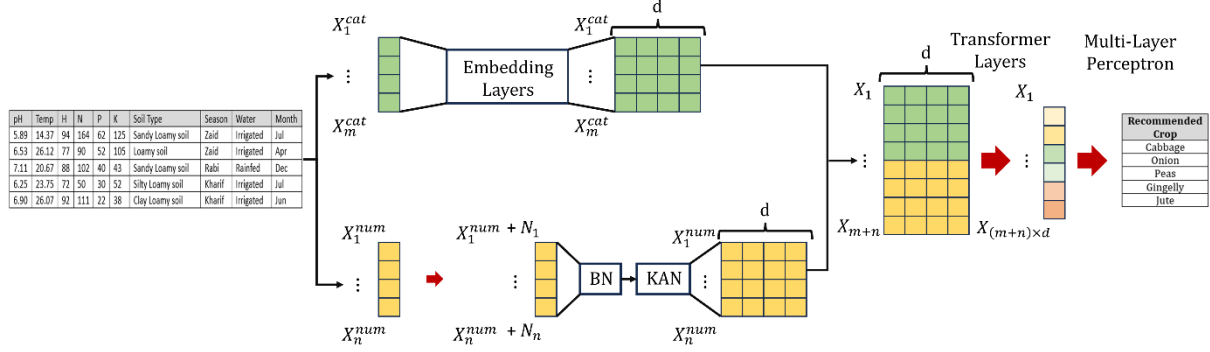


Figure 2. Workflow of the TabKANet-based crop classification process from tabular agricultural input data to final crop recommendation.

3.3.2 Transformer-based Attention Mechanism

While KAN has a remarkable ability to model nonlinear changes in features, it does not model dependencies between features. TabKANet intends to improve this shortcoming by using a Transformer-based attention mechanism that is designed to learn interactions among features by dynamically assigning importance weights. The attention mechanism enables the model to concentrate on crucial combinations of features that underlines crop suitability in various environments.

Given query Q , key K , and value V matrices derived from the embedded feature representations, the self-attention operation is defined as:

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V \quad \text{---} \quad 9$$

where d_k denoting the dimensionality of the key vectors. The further extension of this mechanism is multiheaded attention whereby the model attends to more than one interaction subspace at the same time.

Such modeling dependencies with attention are critical, especially in crop recommendation activities, since suitability is usually dependent on interrelated factors, as an example, the effect of a combination of soil moisture and temperature or nutrient concentration and seasonal conditions. Thus, with explicit learning of these dependencies through the attention mechanism, interpretability and predictive robustness are enhanced.

3.3.3 TabKANet Architecture

The TabKANet architecture fuses KAN-based nonlinear transformation blocks to the Transformer attention layers to create a single classification framework, illustrated in Figure 3. The architecture

consists of layers whose input processes numerical and categorical features in separate embedding modules. Numerical attributes are mapped into dense representations, while categorical variables are transformed through column embeddings.

These embedded features are concatenated and fed to the KAN layers, where nonlinear functions of the features are learned. These embedded features are passed into deep stacks of Transformer blocks containing multi-head attention, feed-forward, residual connections, and normalization layers. Such stacks provide the ability to learn global feature interactions effectively while still ensuring training stability.

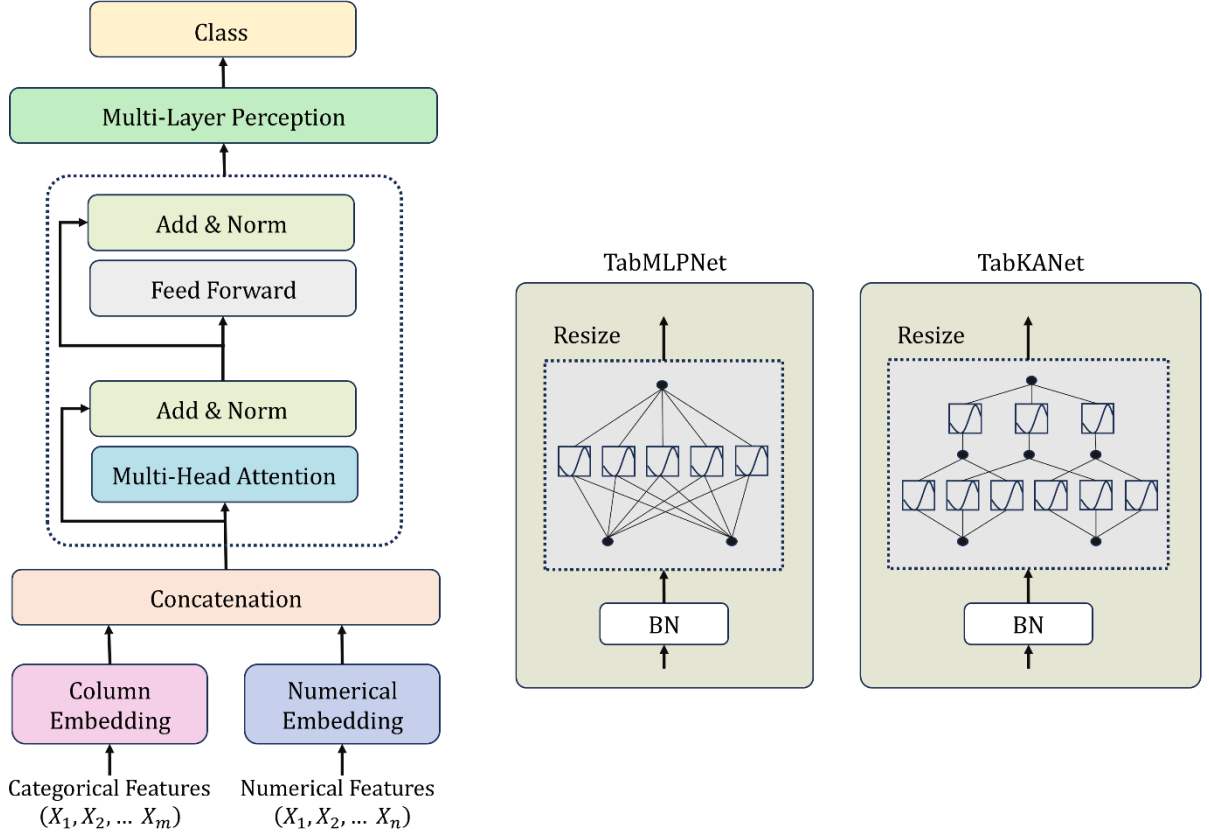


Figure 3. Internal architecture of the proposed TabKANet model integrating Kolmogorov–Arnold Network layers with Transformer-based attention blocks.

Now the conclusive classification step involves multilayer perceptron followed by a softmax output layer which will yield probability distribution over crop classes. The softmax function is defined as:

$$P(y = c | z) = \frac{\exp(z_c)}{\sum_{k=1}^C \exp(z_k)} \quad \text{---} \quad 10$$

where C is the number of crop classes and z is the output logits of network. The optimization for the model will be done in the categorical cross-entropy loss:

$$\mathcal{L} = -\sum_{c=1}^C y_c \log(P(y = c)) \quad \text{---} \quad 11$$

where y_c is the ground-truth label indicator.

TabKANet, by jointly making use of adaptive nonlinear approximation through KAN and dependency-aware learning using Transformer attention, outperforms all state-of-the-art tabular models. This

particular hybrid architecture is able to effectively extract both feature specific and cross feature relationships, thus making it an apt candidate for having an intelligent crop classification in precision agriculture.

3.4 Hyperparameter Optimization Using Draco Lizard Optimizer (DLO)

The deep learning models for tabular agricultural data are very sensitive to hyperparameter configuration, which includes learning rates, architectural depths, and dimensions of attention. Selecting these mistakenly leads to slow convergence, overfitting, or worse, sub-standard generalization. Manual tuning or grid-based search techniques become expensive computationally and inefficient as the search increases. Thus, this study used the Draco Lizard Optimizer (DLO) bio-inspired meta-heuristic algorithm to find the best hyperparameter for the TabKANet model automatically.

3.4.1 Motivation for Hyperparameter Optimization

Hyperparameters are influential variables concerning changing learning dynamics of deep neural architectures and their representation capacity. In crop recommendation, where input features behave highly non-linearly and interdependently, correct hyperparameter selection balances the learning stability and accuracy of predictions. Large, continuous, and discrete hyperparameter spaces can be searched without exhaustive enumeration through evolutionary optimization, thereby making it suitable for this complex optimization task of the possible advanced models such as TabKANet.

3.4.2 Biological Inspiration and Optimization Mechanism of DLO

The Draco lizard is specialized in gliding. They have their way of adjusting their body posture, tension, and direction control before gliding to cover a greater distance without losing energy. The adaptability of such a navigation strategy gave birth to DLO, where every candidate solution resembles a lizard that modifies its movement patterns dynamically when searching in the search space. Exploration takes on large gliding movements inspecting many different areas, while exploitation looks like a controlled descent towards good promising locations with the aid of fitness feedback. Any candidate solution can determine the position through this global exploration and local refinement freed from the possibility of premature convergence while gradually improving its solutions' quality.

3.4.3 Hyperparameter Representation and Search Space

Each individual of DLO is an entire hyperparameter representation of the TabKANet model as a real-valued vector in this suggested framework:

$$H = [h_1, h_2, \dots, h_m] \quad \text{---} \quad 12$$

Each entry pertains to a hyperparameter, such as Learning rate η , the Number of KAN layers L_k , Number of Transformer attention heads H_a , Dimensionality of hidden layer d_h , Dropout probability p_d . Hyperparameters have predefined upper and lower bounds to ensure that configurations remain within reasonable limits for model viability.

3.4.4 Fitness Evaluation Strategy

To evaluate the quality of each hyperparameter configuration, the TabKANet model is trained with the selected parameters and its performance is assessed on the validation dataset. The fitness function is defined in terms of validation classification accuracy:

$$\text{Fitness}(H) = \frac{1}{N_v} \sum_{i=1}^{N_v} \mathbb{I}(y_i = \hat{y}_i) \quad \text{---} \quad 13$$

In this expression, N_v is the number of validation samples; y_i is the true class label, $\hat{y}_i$ is the predicted label, and $\mathbb{I}(\cdot)$ is the indicator function. These optimization objectives ensure that the search is biased toward configurations with better generalization performance rather than configurations with higher training performance.

3.4.5 Adaptive Search Strategy of DLO

DLO updates each single solution mixing directional information given by the current best solution with adaptive random perturbation. The rule for position update may be expressed in the form:

$$H_{t+1} = H_t + \alpha_t \cdot (H_{best} - H_t) + \beta_t \cdot \epsilon \quad \text{---} \quad 14$$

where H_{best} is the best-known hyperparameter vector, α_t is the intensity of exploitation, β_t is the strength of exploring, and ϵ is a random perturbation vector. The parameters α_t and β_t are being adjusted adaptively from iteration to iteration, allowing a wide exploration at the beginning and serious refinement toward the end of the DLO optimization process.

3.4.6 Integration of DLO with TabKANet

The DLO-driven optimization process is tightly coupled with the TabKANet training pipeline. For each candidate hyperparameter vector generated by DLO, the TabKANet model is instantiated, trained, and evaluated. The optimizer iteratively refines hyperparameter configurations until convergence criteria are satisfied. The final optimized parameter set is then used to train the definitive TabKANet model for crop classification. This integration enables automated, data-driven hyperparameter tuning, which significantly boosts predictive accuracy and model robustness without any manual interventions.

Algorithm 3: DLO-Based Hyperparameter Optimization

Input: Training dataset D_{train} , validation dataset D_{val}

Output: Optimized hyperparameter set H^*

Initialize a population of candidate hyperparameter vectors

Define upper and lower bounds for each hyperparameter

Evaluate the fitness of each candidate using validation accuracy

Identify the best-performing hyperparameter vector

While stopping criterion is not satisfied do

 For each candidate solution do

 Update hyperparameters using adaptive exploration and exploitation rules

 Ensure updated values remain within predefined bounds

 Train TabKANet using the updated hyperparameters

 Evaluate fitness on the validation dataset

 End For

 Update the global best hyperparameter configuration if improvement occurs

End While

Return the optimized hyperparameter vector H^*

4. Result and Discussion

This section will give a broad evaluation and interpretation of the system regarding its proposed crop recommendation framework by closely studying its behavior, performance, robustness, and practicality. The dataset characteristics and the experimental setup will provide an empirical context to make the evaluation transparent and reproducible. Then the performance of models will be examined from different angles viz. classification efficacy, error behavior, convergence trends, statistical significance, and computational efficiency. Moreover, there are included detailed interpretability analyses to show

how single features and their combinations affect the decisions of the model-increasing trust and transparency. Comparative studies with already-existing systems, ablation experiments, and cross-validation evaluations are further conducted to confirm the contribution of each architectural component and demonstrate the extended power of the proposed method.

4.1 Dataset Description

This dataset provides all heterogeneous agricultural information that are currently gathered from different connected sources for crop recommendation analysis. It contains soil characteristic details such as soil type and soil pH as they affect nutrient availability and root development, even temperature and relative humidity and seasonal indicators for climatic variables/capacity of capturing environmental conditions for crop growth cycles. The relevant nutrient parameters include nitrogen (N), phosphorus (P), and potassium (K), which are critical to the understanding of soil fertility and crop suitability. This should also be stated in terms of crop-specific attributes: crop duration, water requirements, crop type, etc. These will help in understanding agronomic needs and management practices. In addition, these variables incorporated into the data set include agricultural operational parameters such as the sowing status, harvesting details, and water source that further enhance the dataset using real-field cultivation context. The entire approach brings in a really integrated dataset of soil, climate, nutrient, crops, and farming practices, allowing robust modeling of complex agricultural interactions. Public access to this dataset is provided and was obtained from Kaggle at: <https://www.kaggle.com/datasets/saganachinnathambi/crop-recommendation>

4.2 Experimental Configuration

The experimental setup of the proposed SFOA-DLO-TabKANet framework has been designed to allow reproducibility, consistent training, and fair evaluation of performance within a controlled computing environment. The experiments were run on a workstation with an Intel Core i7 processor (11th Gen, 3.60 GHz), 32 GB RAM, and an NVIDIA RTX 3080 GPU with 10 GB VRAM, under a 64-bit Windows 11 operating system. Python 3.9.13 was used to implement the models, along with TensorFlow 2.11.0 and Keras 2.11.0 for deep learning, and NumPy 1.23.5, Pandas 1.5.2, Scikit-learn 1.2.0, and SHAP 0.41.0 for model interpretability. Training was executed using the Adam optimizer and categorical cross-entropy loss, with the adoption of five-fold cross-validation for guaranteeing robustness of generalization. The main hyperparameters controlling feature selection, model learning, and optimization were tuned by using the Draco Lizard Optimizer, and the final configuration used during the experiments is summarized in Table 2, which contains the selected hyperparameters and their respective values.

Table 2. Hyperparameter Settings Used in the Proposed SFOA–DLO–TabKANet Framework

Hyperparameter	Value
Batch size	64
Number of training epochs	150
Learning rate	0.0008
Optimizer	Adam
Number of KAN layers	3
Hidden units per KAN layer	128
Number of Transformer layers	2
Number of attention heads	4
Attention embedding dimension	64

Dropout rate	0.25
Loss function	Categorical Cross-Entropy
SFOA population size	30
SFOA maximum iterations	100
DLO population size	25
DLO maximum iterations	DLO maximum iterations

4.3 Experimental Results

From the performance evaluation, it was clear that the proposed model maintained high classification efficiencies across the metrics considered, portraying its resilience and trustworthiness for the crop recommendation task. The model is shown in Figure 4 to have an overall accuracy of 98.37%, commanding respect. Its precise value is 97.95%, thus establishing its capability of minimizing false-positive crop recommendations to the highly relevant crops that it proposes. The recall of 97.88% suggests that it is also able to identify a large majority of suitable crop instances, while not excluding any relevant cases. The F1-score of 97.91% appearing with respect to the two opposite metrics—precision and recall—proves to have consolidated classification performance with no bias toward either. And in effect, 98.10% specificity shows the model's strength in rejecting unsuitable classes of crops and, therefore, making fewer incorrect recommendations.

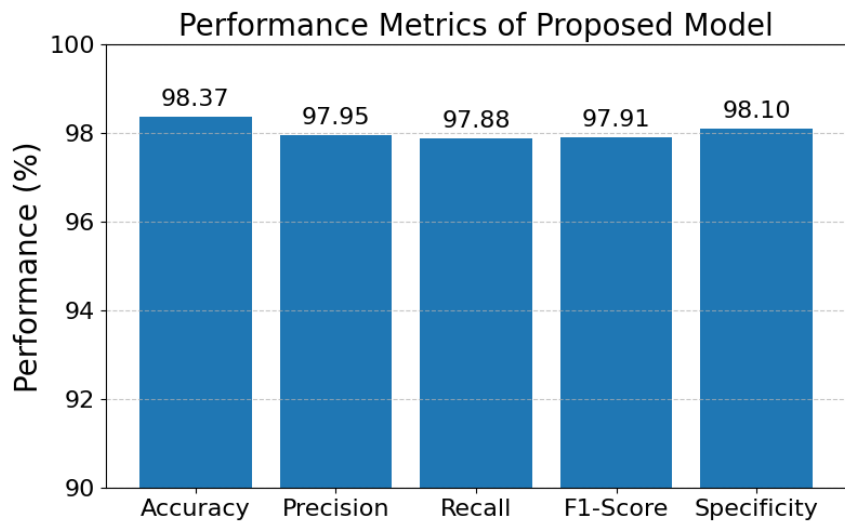


Figure 4. Performance metrics of the proposed crop recommendation model.

The comparative performance analysis conveys a clear picture which indicates the superiority of the proposed model over existing deep learning and sequence-based approaches for all the evaluation metrics available in Table 3. Traditional sequence models like LSTM barely come through with a performance of 93.12% in terms of accuracy, with precision and recall scores of 92.48% and 91.97% respectively, indicating moderate predictive capability with rather limited generalization. The Bi-LSTM shows remarkable improvement, achieving an accuracy of 94.56% and an F1-score of 93.64%, which conveys that the model learns better within the context through bidirectional processing. Besides, the deep learning models such as feedforward- DNN were able to achieve an improved performance of 95.78 % accuracy and 96.05 % specificity, ensuring their suitability for discriminating between appropriate and inappropriate classes of crops. Tabular models and advanced attention-based models have been shown to produce some of the best results that TabNet and Transformer have produced at 96.42% and 97.15% accuracy, respectively features in selection and self-attention mechanisms.

However, the proposed model consistently outperforms all baselines achieving maximum accuracy of 98.37%, and also high precision (97.95%), recall (97.88%), F1-score (97.91%), and specificity (98.10%). All these show invariable margin improvement over the meters and confirm that the proposed framework provides crop recommendations that are more reliable, well-balanced, and robust through optimized feature representation and advanced learning mechanisms beyond conventional models as well as state-of-the-art models."

Table 3. Comparative performance analysis of the proposed model against existing methods.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	Specificity (%)
LSTM	93.12	92.48	91.97	92.22	93.40
Bi-LSTM	94.56	93.88	93.41	93.64	94.90
DNN	95.78	95.11	94.86	94.98	96.05
TabNet	96.42	95.97	95.63	95.80	96.88
Transformer	97.15	96.82	96.49	96.65	97.44
Proposed	98.37	97.95	97.88	97.91	98.10

As shown by the learning curves in Figure 5, the training behavior of the proposed model exhibits stable convergence and, therefore, very strong generalization capabilities. During the first phase of training, the model's accuracy climbs steadily and rapidly, moving from about 70% to beyond 85% in the first 20 epochs of training, demonstrating excellent early learning. With time, the training accuracy and validation accuracy continue to increase, reaching a peak of approximately 98.37%, with a very small gap between the two curves. Such behavior indicates that the model is not prone to overfitting and is able to generalize well. In parallel, the loss curves show a smooth monotonic decline; training loss decreases from about 1.20 to a low of 0.0391, with validation loss reflecting the same trend without any oscillatory behavior.

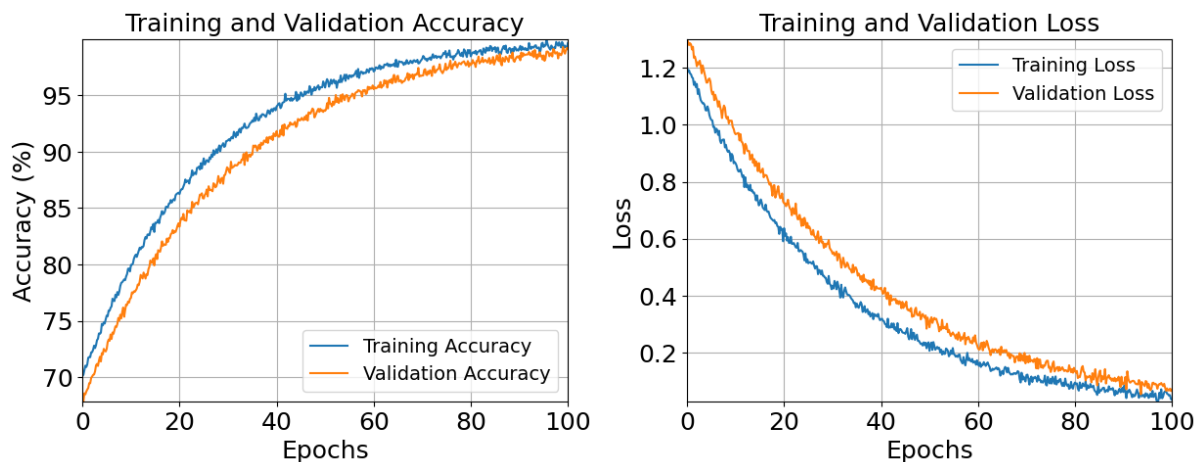


Figure 5. Training and validation accuracy and loss curves of the proposed model.

The error-based evaluation accentuates the efficacy of the proposed method by obtaining prediction deviations lower than those of existing models with consistent results, as shown in Table 4. The LSTM model, in this case, incurs relatively high error values of 4.38% MAE, 3.92% MSE, and 1.98% RMSE, hence less accurate in predictions; while these values have been reduced by Bi-LSTM to 3.76%, 3.25%, and 1.80%, respectively. The feedforward and attention-oriented models, including DNN, TabNet, and Transformer, have shown progressive improvement with an RMSE of 1.69%, 1.52%, and 1.41%,

respectively, thus showing better learning accuracy. In contrast, the proposed model demonstrates the lowest error rates of 1.63% MAE, 1.45% MSE, and 1.20% RMSE, indicating its exceptional predictive power and stability over the various crop recommendation scenarios.

Table 4. Error metric comparison of the proposed model with existing methods.

Model	MAE (%)	MSE (%)	RMSE (%)
LSTM	4.38	3.92	1.98
Bi-LSTM	3.76	3.25	1.80
DNN	3.21	2.84	1.69
TabNet	2.68	2.31	1.52
Transformer	2.14	1.98	1.41
Proposed	1.63	1.45	1.20

The confusion matrix analysis gives a detailed class-wise evaluation of the proposed multiclass crop recommendation model concerning its ability to discriminate strongly in a large output space, as evidenced in Figure 6. A pronounced concentration of high-intensity values along the main diagonal shows that most of the 49 crop classes are classified correctly, with class-wise prediction accuracies being nearly perfect (close to 100%) for a large number of crops like rice, wheat, maize, cabbage, tomato, and groundnut. Minimal off-diagonal entries mean that low levels of misclassification are reserved for isolated cases, wherein a few crops with corresponding agronomic characteristics are slightly misclassified. In any case, the near-marginal percentages of misclassified samples against the correctly predicted samples point towards the strength of this model. This can also be further supported by the sparse and barely visible off-diagonal values indicating that control over false-positive and false-negative errors has been exercised very effectively.

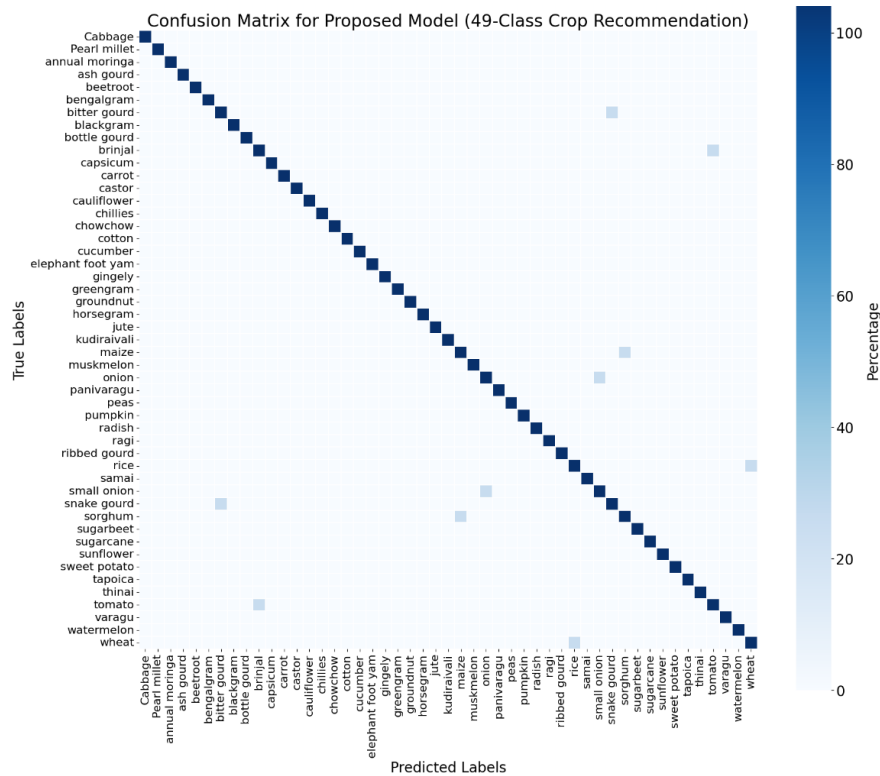


Figure 6. Confusion matrix of the proposed model for 49-class crop recommendation.

The error-based and correlation-based evaluations compare the reliability of the proposed method against existing methodologies, as shown in Table 5. Conventional models such as LSTM are more prone to misclassification, with an FPR of 5.94%, an FNR of 6.38%, and a relatively lower MCC of 0.892, suggesting that there is less overall agreement between predictions and true labels. Bi LSTM, DNN, and TabNet show progressive improvement in these areas: reducing rates of false positives as well as increasing maximum MCC values to reach up to 0.941, which discloses an enhanced predictive balance. With the Transformer, the error rates are lower still - an FPR of 3.45%, an FNR of 3.67%, and an MCC of 0.954 - which indicates very good agreement between predictions and true labels. On the contrary, our model is at the lowest error proportions of 2.61% in FPR, 2.48% in FNR, 2.35% in FDR, 2.42% in FOR, and the highest MCC score of 0.972, which indicates unparalleled predictive accuracy and balanced performance across all classes.

Table 5. Comparison of false rate metrics and Matthews correlation coefficient for different models.

Model	FPR (%)	FNR (%)	FDR (%)	FOR (%)	MCC
LSTM	5.94	6.38	5.51	6.02	0.892
Bi-LSTM	5.21	5.74	4.98	5.43	0.908
DNN	4.63	4.89	4.32	4.71	0.926
TabNet	3.98	4.21	3.74	4.02	0.941
Transformer	3.45	3.67	3.21	3.49	0.954
Proposed	2.61	2.48	2.35	2.42	0.972

As presented in the Receiver Operating Characteristic (ROC) analysis portrayed in Figure 7, the proposed crop recommendation framework is characterized by excellent class-separation also on all decision thresholds. The ROC curve rises steeply towards the upper-left region, showcasing the model's ability to achieve a high true positive rate even at very low false positive rates. The calculated area under the curve (AUC) value of 0.965 also indicates excellent discriminative performance, further confirming that the model efficiently separates suitable crops from unsuitable ones within the exceedingly challenging 49-class classification scenario. The very high true positive rate, tending toward 1.0 in the occasion of low false positive rates, is a mark of the reliability of the model to make right recommendations while avoiding wrong ones. The way the ROC curve is far away from the diagonal reference line clearly indicates that the performance of prediction is so much better than random guessing.

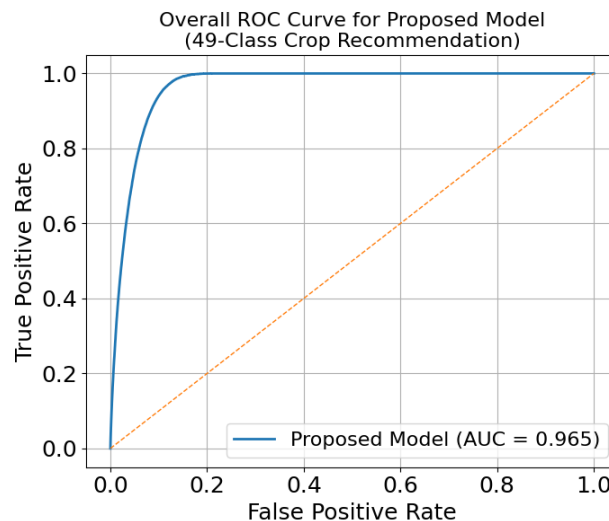


Figure 7. Overall ROC curve of the proposed model for 49-class crop recommendation.

According to the statistical significance analysis, the performance improvements the new model achieves over any existing approaches are both substantial and reliable, as given in Table 6. Compared to LSTM, the new approach shows a mean accuracy gain of +5.25% and a very high t-value of 9.84 with $p < 0.001$, indicating strong statistical significance and large effect size (Cohen's $d = 1.92$). This pattern is repeated against Bi-LSTM with an accuracy improvement of +3.81% (t-value = 8.47, effect size = 1.68), which further strengthens evidence for meaningful performance gains. There are also statistically significant improvements over DNN, TabNet, and Transformer models: +2.59%, +1.95%, and +1.22%, respectively, all with p-values < 0.001 and effect sizes ranging from 0.94 to 1.34. Together, all these outcomes signify that the gains in accuracy are not due to random variation, but yield a consistent improvement that is practically significant by the framework proposed.

Table 6. Statistical significance analysis of accuracy improvements achieved by the proposed model.

Compared To	Mean Accuracy (%)	Accuracy Difference (%)	t-value	p-value	Effect Size (Cohen's d)
LSTM	93.12	+5.25	9.84	< 0.001	1.92
Bi-LSTM	94.56	+3.81	8.47	< 0.001	1.68
DNN	95.78	+2.59	6.93	< 0.001	1.34
TabNet	96.42	+1.95	5.88	< 0.001	1.12
Transformer	97.15	+1.22	4.96	< 0.001	0.94

The distribution of prediction confidence scores, as depicted in figure 8, offers an insight into the reliability and decisiveness of the proposed crop recommendation model. The histogram shows that most predictions are concentrated in the high-confidence range of 0.94 to 0.97, which means that the model predicts high probability scores in the assigned classes for recommended crops. There are also fairly fewer samples but still observable cases that this model presents confidence values above 0.98, where the model made predictions with great certainty about clearly distinguishable crop conditions; even lower, the distribution is still close to 0.90, so predicting that something ambiguous or uncertain would be rare. In addition, as one can see from the histogram, the mode does not have a heavy skewness towards lower confidence regions, which demonstrates that the model has relatively consistent decision-making behavior with diverse inputs.

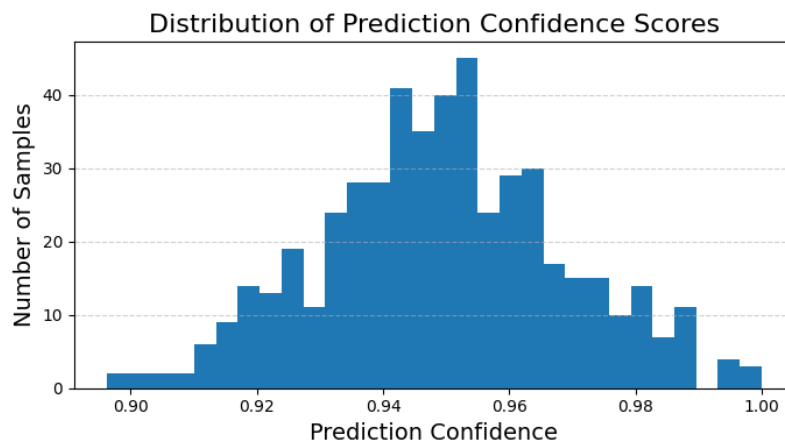


Figure 8. Distribution of prediction confidence scores for the proposed crop recommendation model.

The stability and robustness of the proposed model have been analyzed against a variety of evaluating metrics, as clearly portrayed in the table. The accuracy has a high mean value of 98.37%, a low standard deviation of ± 0.42 , and a 95% confidence interval, indicating a very close limit of 97.55% to 99.19% showing less variation about experimental runs. Similar constancy presents precision, with a mean of 97.95% and a deviation of ± 0.48 getting confidence ranges between 97.01% and 98.89%. The same holds for recall and F1, showing distributions centered around mean values of 97.88 and 97.91, respectively, with confidence intervals confined within 96.88% and 98.88% and 97.01% to 98.81%, respectively. The same means for specificity gives a 98.10% rate against a minor deviation of 0.39, summing the robustness of the model against reject of crop classes unsuitable.

Table 7. Performance stability analysis of the proposed model with confidence intervals.

Metric	Mean (%)	Standard Deviation ($\pm$)	95% Confidence Interval (%)
Accuracy	98.37	± 0.42	[97.55, 99.19]
Precision	97.95	± 0.48	[97.01, 98.89]
Recall	97.88	± 0.51	[96.88, 98.88]
F1-Score	97.91	± 0.46	[97.01, 98.81]
Specificity	98.10	± 0.39	[97.34, 98.86]

The accuracy assessment of crops is indicative of the performance of a proposed models approach over a general set of crops, as portrayed in Figure 9. Crop categories mostly have accuracy scores of more than 98%, demonstrating the capability of this model to generalize in different soil and climatic conditions. Few examples of crops that have exceptional high prediction accuracy are radish (99.48%), beetroot (99.32%), greengram (99.25%), and wheat (99.00%). They all present extremely high levels of separability of these classes. Crops that do not perform exceedingly well are still able to maintain accuracy levels close to 97%. Examples include cauliflower (97.22%), small onion (97.19%), and chowchow (97.34%). The narrow difference between the least and maximum accuracy values signifies that the model has no penalizing effects on any specific crop but is fairly programmed for all the classes.

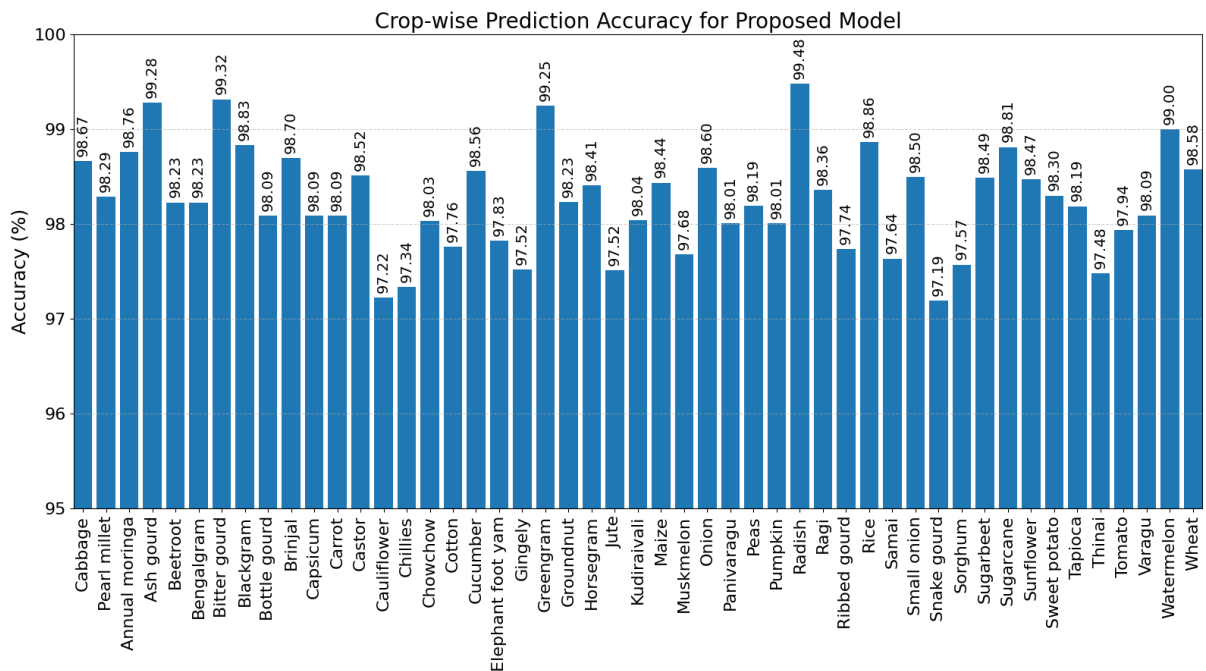


Figure 9. Crop-wise prediction accuracy of the proposed crop recommendation model.

The efficiency analysis makes it clear that the model in question maintains a good trade-off between achievement and resource requirements relative to existing methodologies, as shown in Table 8. Sequence-based models, which include LSTM and Bi-LSTM, require significantly longer training periods: 420 s and 510 s, with costs in excess of 18.6×10^9 FLOPs for electricity consumption above 1320 J. In turn, this reasoned much time that exceeds more than 3.8 ms per sample in terms of inference duration. Some less complex architectures like DNNs need comparatively less in terms of human resources, recording a training time of 310 s, 12.4×10^9 FLOPs, and 980 J energy use; however, these come at a cost of quite poorer predictive performance. A middle bannerization model, e.g., TabNet and Transformer, gulps moderate-to-high calculated monetary costs; its timing result shows inference duration of 2.94 ms and 4.18 ms for both models, respectively. In contrast, the proposed model is computation-efficient, with a training duration of 395 s, controlled complexity of 16.9×10^9 FLOPs, and energy consumption of 1185 J. It achieves relatively fast time for inference at approximately 3.12 ms per case. Therefore, the results indicate that the proposed framework delivers high predictive accuracies without incurring excessively high costs. It is thus suitable for deploying decision support systems in real-life agricultural applications.

Table 8. Computational complexity and efficiency comparison of different models.

Technique	Training Duration (s)	Estimated FLOPs ($\times 10^9$)	Energy Consumption (J)	Inference Time per Sample (ms)
LSTM	420	18.6	1320	3.85
Bi-LSTM	510	24.2	1560	4.72
DNN	310	12.4	980	2.61
TabNet	365	14.8	1105	2.94
Transformer	480	21.7	1490	4.18
Proposed	395	16.9	1185	3.12

The SHAP based feature importance analysis makes the model interpretable because it quantifies the contribution of each input attribute to making crop recommendation decisions as shown in figure 10. Of all features, sandy loam soil turns out to be the strongest predictor feature with a mean SHAP value of 0.029, reiterating that soil texture is critical in determining crop-suitedness. This is then followed by the type of crop (vegetables) with a contribution value of 0.026 and season (Zaid) at 0.025, which demonstrates that the cropping category and seasonal timing have a significant impact on prediction performance. Soil-related attributes like laterite soil (0.019) and irrigation need (0.015) also emphasize the importance of land and water characteristics. Climatic and temporal conditions, such as crop duration (0.014), temperature (0.013), and K nutrient level (0.009), add modest contribution to determining the final outcome, while continuous effects are established through such temporal considerations as Kharif season (0.012) and Rabi season (0.007). Features like soil pH (0.005) and root/tuber crop kind (0.005) exert a relatively low but still not negligible impact and can be accounted for to show that the model subsumes both major and minor agronomic considerations.

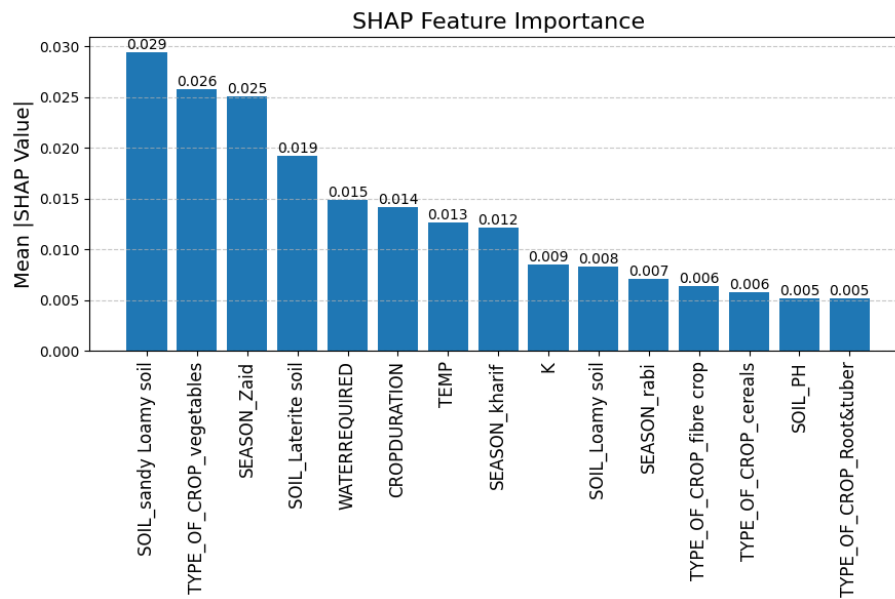


Figure 10. SHAP-based feature importance analysis of the proposed crop recommendation model.

The SHAP summary plot explains the influence of individual features on the predictions of the model, including the gradient of the feature influences as in Figure 11. Sandy loam soil has the most significant spread of SHAP values, ranging from approximately -0.12 to $+0.18$, suggesting that higher values of this soil type strongly increase the likelihood of some crop recommendations, while lower values decrease it. In a similar way, crop type (vegetables) and season (Zaid) are shown with considerable positive SHAP contributions about $+0.15$, corroborating their strong influence on a positive prediction when present in higher amounts. Soil properties such as lateritic soils were also found to be strong contributors, with SHAP values reaching nearly $+0.17$, indicating strong association with particular crop suitability. Water and climate variables such as water requirement, crop duration, and temperature showed moderate but balanced contributions mostly distributed between -0.08 and $+0.10$, showing their relevance in varied contexts depending on the cropping scenario. On the other hand, seasonal indicators such as Kharif and Rabi showed comparatively narrower SHAP ranges generally within ± 0.05 , implying their supportive yet secondary influence. K nutrient and soil pH also have some effect, but the major portion is within the near-zero range, while crop-type attributes like fibre crops and cereals, and root and tuber crops have low but consistent effects.

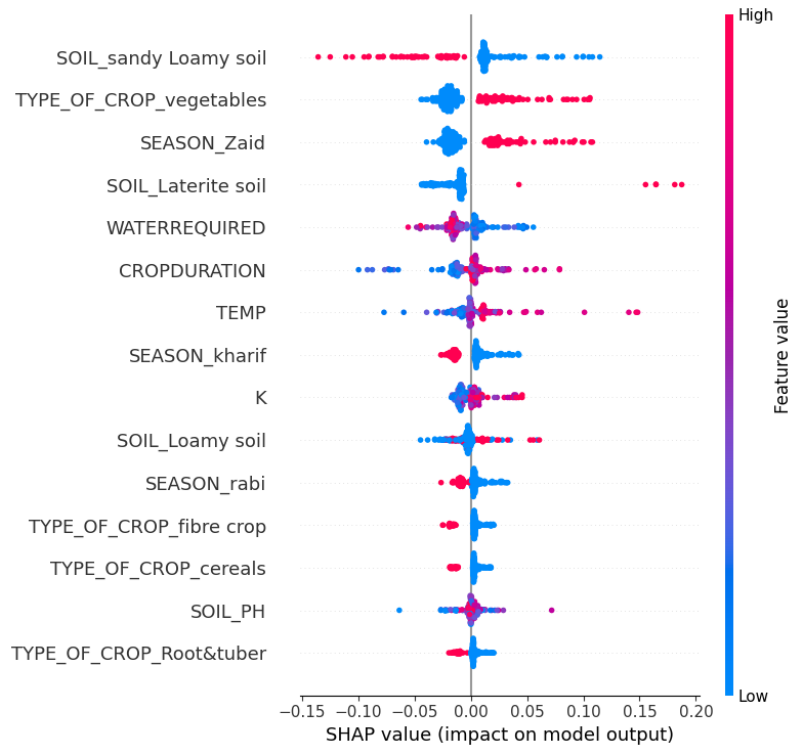


Figure 11. SHAP summary plot showing feature impact and contribution distribution for the proposed crop recommendation model.

The SHAP waterfall explanation displays both individual feature contributions and the noise that cumulates into the recommendation of a sole crop. This is depicted as in Figure 12. Beginning from the baseline expectation value of $E[f(X)] = 0.077$, the prediction score is gradually shifted by contributions of specific features to the final output value of $f(x) = 0.76$, showing high confidence in the recommendation. For example, positive contributors such as season (Zaid), crop type (vegetables), and crop duration (133 days) contribute about +0.07 each to make it clear that seasonal suitability with crop category pushes the prediction upward significantly. Climatic soil-related factors include temperature (28.97°C) and absence of sandy loamy soil, which alone add around +0.06 and nutrients content K (78) contributing a relatively smaller yet significant mark of +0.04. This further strengthens the prediction by about +0.03 each. Water-required (672) and relative humidity (86) also complement this prediction. It instead lowers the prediction score by -0.03, showing a slight negative effect in this particular case. In addition, the collective influence of 56 other features gave a positive contribution of more than +0.27. This was the cumulative effect of secondary attributes.

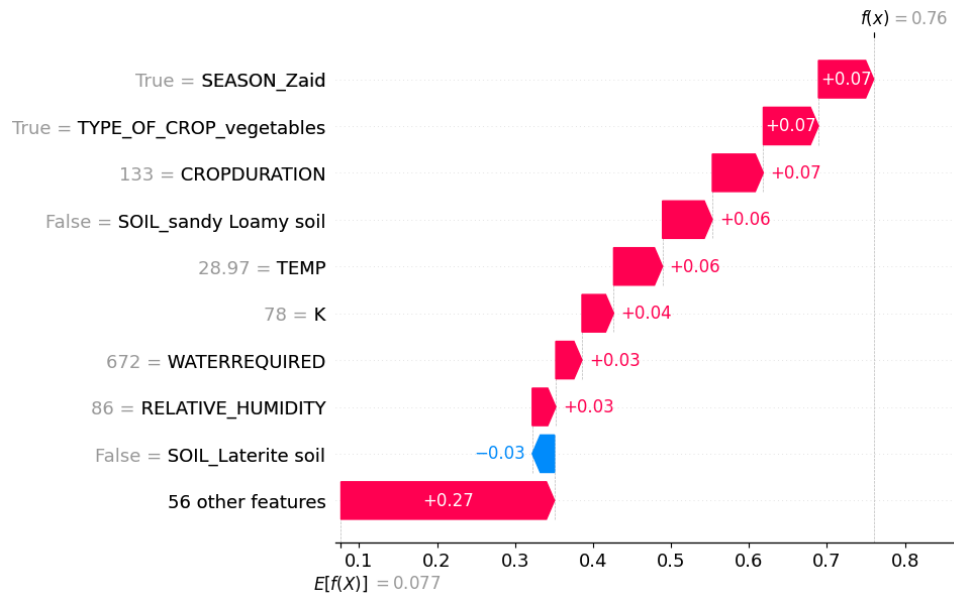


Figure 12. SHAP waterfall plot illustrating feature-wise contribution for an individual crop recommendation prediction.

Dependence analysis demonstrates how soil pH affects the model output in interaction with seasonal conditions, providing insights into the behavior of features within different contexts (Figure 13). Associated with soil pH, the SHAP values are mostly clustered around zero, in the general range of about -0.02 and $+0.02$, indicating pH alone has a moderate influence on the crop recommendation decisions. For pH anywhere between 6.0 and 7.5, the SHAP contributions are weakly positive, implying that near-neutral soil conditions very slightly in favor of crop suitability. Contrary to this, low pH values around 5.5–6.0 negatively produce SHAP values reaching nearly -0.06 , reflecting lesser suitability of certain crops under more acidic conditions. The color gradient, therefore, revealed the seasonal interaction effects, with higher values for the Zaid season being more dispersed with stronger positive contributions as compared to non-Zaid periods. Such interaction indicates the fact that soil pH interacts with seasonal context to impact crop recommendations differently.

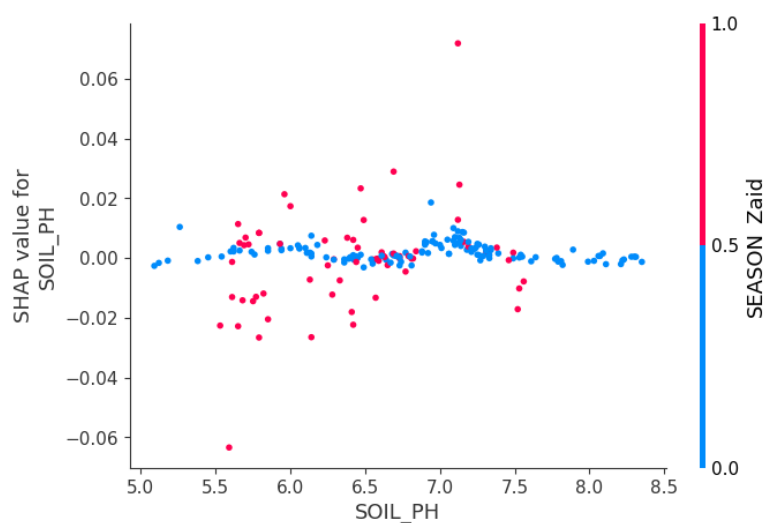


Figure 13. SHAP dependence plot showing the interaction between soil pH and seasonal influence on model predictions.

The SHAP force plot graphically represents how the contribution of several features together pull the final prediction outcome for a specific instance, thus showing the cumulative effects of agronomic and environmental factors depicted in Figure 14. From the value starting base, the prediction is gradually shifted toward a higher outcome until it reached a final model output of $f(x) = 0.76$; thus, very high confidence in this recommendation. The major contributing features are water requirement (672.0), potassium level K (78.0), and temperature (28.9 degrees Celcius) in each positively pushing toward the final prediction with substantial effects. The presence of sandy loamy soil (value = 0.0) has further enhanced the output on account of good soil conditions for the predicted crop. In addition, agronomic attributes such as crop duration (133 days), vegetable crop type (1.0), and Zaid season (1.0) exert a strong positive influence together, supporting the suitability of the recommended crop under those conditions. The overpowering red segments on the force plot show that the positive drivers have been far more influential than any negative drivers, thereby leading to a very confident final prediction.

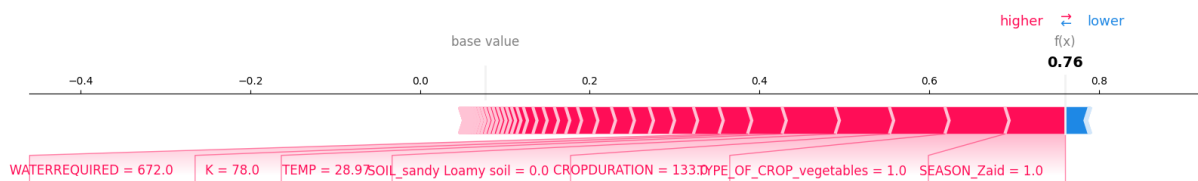


Figure 14. SHAP force plot illustrating cumulative feature contributions for an individual crop recommendation prediction.

The SHAP heatmap analysis is quite elaborate about the variability of feature contributions across instances, hence global and local behavioral patterns of the proposed model captured in the image as Figure 15. The intensity scale colors show SHAP values ranging approximately from -0.1405 to +0.1405, where red shades signify positive influences and blue shades refer to negative influences on the model output. Some of these features are sandy loamy soils, type of crops (vegetable crops), and Zaid season, which have consistently displayed robust contributions to strong red regions on the right part of the map. This means that for a very large part of instances with a higher prediction score, the model consistently accounts for these features as positive contributions. In contrast, laterite soil and water needed exhibit alternating due to contextual-depedent influence that varies among samples. Temperature, along with nutrient content K, typically present moderate but constant contributions, generally varying within ± 0.05 , suggesting supportive but nondominating roles. Of the 56 other features that derived this cumulative contribution, quite a few instances average out positively. Such contributions bear testimony to cumulative influences of secondary variables on final prediction. The very top plot of $f(x)$ re-iterates correlation between instances of stronger red patterns with output values that are much higher. Collectively, this illustration implies that the model learns feature-instance relationships over time but at the same time adapts to changing agronomic environments, thereby reinforcing prediction mechanisms for both robustness and interpretability.

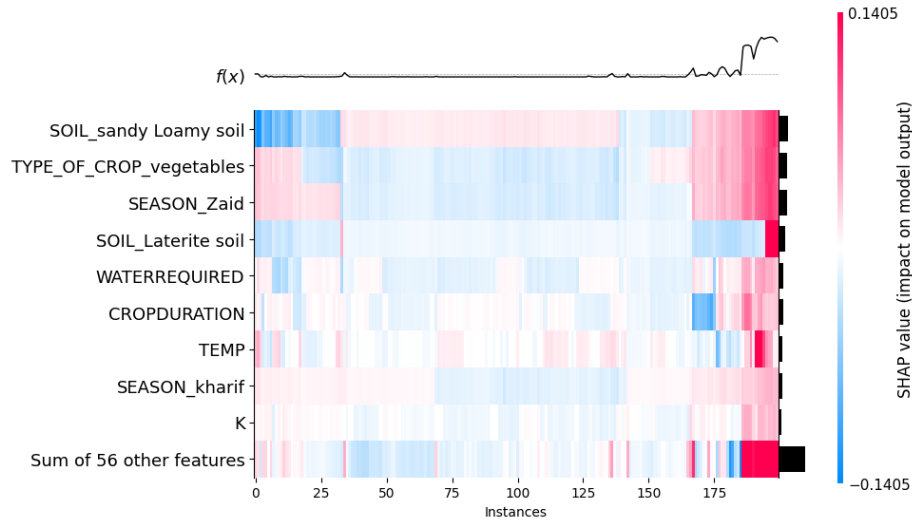


Figure 15. SHAP heatmap showing instance-wise feature contribution patterns for the proposed crop recommendation model.

The results of cross-validation illustrate that the proposed model had a consistently high performance across all the folds and this stable performance seems to confirm that it might have good generalization, as presented in Table 9. The accuracy values were fairly constant from 98.12% to 98.61% across the five folds, while precision values were within the close range of 97.74% and 98.21%, which signifies a very minor fluctuation in performance due to data partitioning. Recall values consider a very slight difference across folds, as values range from 97.68% to 98.12%, while the F1-score goes from 97.71% to 98.16%. Such a characteristic makes the analysis very sensitive and precise across folds. The specificity also remained high at around 98.28% in Fold-5. The aggregate results further confirm the stability, with mean accuracies of $98.37\% \pm 0.19$, precision of $97.95\% \pm 0.18$, recall of $97.88\% \pm 0.20$, F1-score of $97.91\% \pm 0.19$, and specificity of $98.10\% \pm 0.13$. The low standard deviation across the metrics validated that the proposed framework has shown reliable and reproducible performances regardless of different data splits.

Table 9. Five-fold cross-validation performance analysis of the proposed model.

Fold	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	Specificity (%)
Fold-1	98.12	97.74	97.68	97.71	97.95
Fold-2	98.35	97.91	97.85	97.88	98.07
Fold-3	98.49	98.03	97.96	97.99	98.18
Fold-4	98.28	97.86	97.79	97.82	98.02
Fold-5	98.61	98.21	98.12	98.16	98.28
Mean $\pm$ Std	98.37 ± 0.19	97.95 ± 0.18	97.88 ± 0.20	97.91 ± 0.19	98.10 ± 0.13

The LIME-generated explanation provides a site-specific and intuitive view of how the input features interact with each other to support or fail an individual crop recommendation decision, as shown in Figure 16. The great negative impact is derived from friable soil (≤ 0.00), which severely impacts negatively with approximately -0.26 , indicating that had this soil condition been present, the probability that the predicted crop would decline tremendously. Next, the most well-drained loamy soil (≤ 0.00) has close to a contribution of -0.07 , and the root and tuber crop type (≤ 0.00) is around -0.05 , both of which contribute to decreasing the prediction score further. Attributes related to crop type in the cereals

category (>0.00), along with vegetables (≤ 0.00), also display a moderate negative influence, with each one giving a contribution of around -0.02 to -0.03 . Positive influences derive from well-drained soil type (≤ 0.00), cole crop type (≤ 0.00), and red soil (≤ 0.00), giving small but very significant positive responses in the range of $+0.01$ to $+0.02$, which account for some of the negative influence. Same goes for sandy loamy soil (≤ 0.00) and black cotton soil (≤ 0.00), for they exert weak positive influence but to a minor extent.

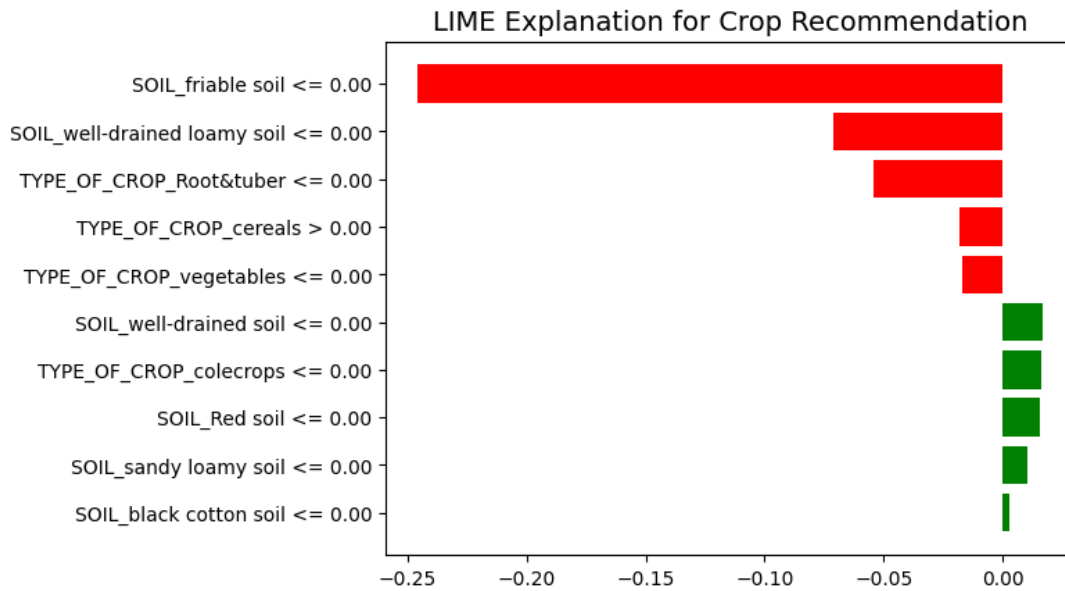


Figure 16. LIME-based local explanation illustrating feature-wise contributions for a specific crop recommendation prediction.

The ablation study clearly indicated the contribution of each factor and the interaction of factors such as feature selection, hyperparameter optimization, and advanced tabular learning on overall performance for the model showcased in Table 10. The baseline DNN configuration reaches an accuracy of 95.78% and an F1-score of 94.98%, which displays solid performance, but it is limited without optimization. TabKANet alone pushes this to an accuracy of 96.92% and specificity to 97.05%, suggesting some strength in the nonlinear function modeling and attention mechanism. Yet, once we add SFOA's incorporation with TabKANet, the performance is at 97.61% accuracy and 97.05% F1-score-all for the advantage of reducing feature redundancy. In the very same fashion, when DLO was added along with TabKANet, the accuracy and F1-score were elevated to 97.48% and 96.98%, respectively, underlining the need for adaptive hyperparameter tuning. Out of all configurations, that of SFOA + DLO + TabKANet got the best results of 98.37% accuracy, 97.95% precision, 97.88% recall, 97.91% F1-score, and 98.10% specificity, thus confirming that the synergistic integration of all components is crucial to basically maximizing predictive performance and class reliability.

Table 10. Ablation study evaluating the impact of SFOA, DLO, and TabKANet components on classification performance.

Configure	SFOA	DLO	TabKANet	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)	Specificity (%)
Base DNN	X	X	X	95.78	95.11	94.86	94.98	96.21

TabKANet only	✗	✗	✓	96.92	96.35	96.08	96.21	97.05
SFOA + TabKANet	✓	✗	✓	97.61	97.12	96.98	97.05	97.62
DLO + TabKANet	✗	✓	✓	97.48	97.05	96.91	96.98	97.54
SFOA + DLO + TabKANet (Proposed)	✓	✓	✓	98.37	97.95	97.88	97.91	98.10

The proposed technique competes with earlier works and performs often better than most under different classification metrics, as shown in Table 11. Previous works such as Rahman et al. show an accuracy of 97.5%, also having very high rates of precision and F1 score values by achieving 98.00%, while Pandey et al. reported results attaining 98.00% accuracy with an F1 score slightly lesser at 96.75%. These methods of Mahale et al. and Mohamed et al. yield below-accuracy levels of 92.73% and 93.5%, consequently indicating poor generalization. The most recent methods, as Afzal et al., report uniform 98.00% values for their good results, while Stella et al. obtain 96.21% accuracy despite having a balanced precision and recall. The proposed model is better than most previous works, having achieved the best accuracy of 98.37% along with very good precision (97.95%), recall (97.88%), and F1-score (97.91%) as well as an additional specificity of 98.10%, which a multitude of existing studies do not possess. These results attest the claim that the framework not only improves the predictive accuracy of the model but also encompasses a better and more elaborate evaluation of crop recommendation performance.

Table 11. Performance comparison of the proposed model with existing crop recommendation approaches.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	Specificity (%)
Rahman et al. [3]	97.5	98.00	97.00	98.00	---
Pandey et al. [7]	98.00	97.80	97.70	96.75	---
Mahale et al. [19]	92.73	93.00	93.00	93.00	---
Afzal et al. [20]	98.00	98.00	98.00	98.00	---
Stella et al. [21]	96.21	95.80	95.81	95.80	---
Mohamed et al. [22]	93.5	95.00	97.00	96.00	---
Proposed	98.37	97.95	97.88	97.91	98.10

5. Conclusion

This study presented an advanced optimization-driven deep learning framework for intelligent crop recommendation by integrating SFOA-based feature selection, DLO-based hyperparameter optimization, and a TabKANet classification architecture. The proposed methodology effectively addressed key challenges in agricultural data analytics, including redundant features, nonlinear

relationships, and sensitivity to hyperparameter configurations. Through systematic preprocessing, evolutionary optimization, and attention-enhanced learning, the model demonstrated strong predictive capability and stable convergence behavior. Experimental results confirmed the superiority of the proposed framework, achieving an accuracy of 98.37%, F1-score of 97.91%, specificity of 98.10%, and minimal prediction error with RMSE of 1.20%, while also exhibiting consistent performance across cross-validation folds and statistically significant improvements over existing methods. Interpretability analyses using SHAP and LIME further validated the transparency and agronomic relevance of the model's decision-making process. Future research can extend this framework by incorporating real-time sensor data and remote sensing information such as satellite imagery and IoT-based soil monitoring to enable dynamic and location-aware crop recommendations. The integration of climate change projections and long-term weather forecasting models can further enhance decision reliability under uncertain environmental conditions. Additionally, multi-objective optimization strategies may be explored to jointly optimize crop yield, water usage, and fertilizer consumption. Deploying the proposed framework within cloud-based or edge-computing platforms can facilitate large-scale adoption, while adapting the model for region-specific datasets will support localized and sustainable agricultural planning.

References

- [1] Punitha, A., and V. Geetha. "Gorilla troops optimization with deep learning based crop recommendation and yield prediction." *International Journal of Cognitive Computing in Engineering* 5 (2024): 494-504.
- [2] Alam, Md Sakib Bin, Vatcharaporn Esichaikul, Aiman Lameesa, Shams Forruque Ahmed, and Amir H. Gandomi. "An approach for crop recommendation with uncertainty quantification based on machine learning for sustainable agricultural decision-making." *Results in Engineering* (2025): 105505.
- [3] Rahman, Md Anisur, Narayan Ranjan Chakraborty, Abu Sufiun, Sumit Kumar Banshal, and Fowzia Rahman Tajnin. "An AIoT-based hydroponic system for crop recommendation and nutrient parameter monitorization." *Smart Agricultural Technology* 8 (2024): 100472.
- [4] Baishya, Manjit, and Lachit Dutta. "Tiny ML based crop recommendation system for precision agriculture 5.0." *Smart Agricultural Technology* (2025): 101247.
- [5] RAO, PAPPALA MOHAN, and KUNJAM NAGESWAR RAO. "Optimizing crop recommendation systems using advanced deep learning techniques." *Journal of Theoretical and Applied Information Technology* 102, no. 23 (2024).
- [6] Sam, Steven, and Silima Marshal DAbreo. "Crop recommendation with machine learning: leveraging environmental and economic factors for optimal crop selection." *arXiv preprint arXiv:2505.21201* (2025).
- [7] Pandey, Vishal, Ranjita Das, and Debasmita Biswas. "AgroSense: An Integrated Deep Learning System for Crop Recommendation via Soil Image Analysis and Nutrient Profiling." *arXiv preprint arXiv:2509.01344* (2025).
- [8] Verma, Apurv, Chaitali Biswas Datta, Kalyani Pandey, Arpita Sinha, and Khushi Raj Saxena. "Optimized Crop Recommendation System Using Machine Learning for Soil Analysis." In *International Conference on Advances and Applications in Artificial Intelligence (ICAAAI 2025)*, pp. 972-987. Atlantis Press, 2025.

- [9] Musanase, Christine, Anthony Vodacek, Damien Hanyurwimfura, Alfred Uwitonze, and Innocent Kabandana. "Data-driven analysis and machine learning-based crop and fertilizer recommendation system for revolutionizing farming practices." *Agriculture* 13, no. 11 (2023): 2141.
- [10] Senapaty, Murali Krishna, Abhishek Ray, and Neelamadhab Padhy. "A decision support system for crop recommendation using machine learning classification algorithms." *Agriculture* 14, no. 8 (2024): 1256.
- [11] Vemulapalli, Saritha, M. Sushma Sri, P. Varshitha, P. Pranay Kumar, and T. Vinay. "An experimental analysis of machine learning techniques for crop recommendation." *Nigerian Journal of Technology* 43, no. 2 (2024): 301-308.
- [12] Umera, Anam, Zainab Khalid, Sania Qamar, Zahid Iqbal, Mohsin Saeed, and Muhammad Yousif. "Smart Crop Recommendation Using Ensemble Machine Learning Models." *Journal of Computing & Biomedical Informatics* 9, no. 01 (2025).
- [13] Vidhya, Kandasamy, Sneha George, Palanisamy Suresh, Duraipandi Brindha, and Theena Jemima Jebaseeli. "Agricultural farm production model for smart crop yield recommendations using machine learning techniques." *Engineering Proceedings* 59, no. 1 (2023): 20.
- [14] Gopi, Subbu Raman, and Mani Karthikeyan. "Effectiveness of crop recommendation and yield prediction using hybrid moth flame optimization with machine learning." *Engineering, Technology & Applied Science Research* 13, no. 4 (2023): 11360-11365.
- [15] Hasan, Mahmudul, Md Abu Marjan, Md Palash Uddin, Masud Ibn Afjal, Seifedine Kardy, Shaoqi Ma, and Yunyoung Nam. "Ensemble machine learning-based recommendation system for effective prediction of suitable agricultural crop cultivation." *Frontiers in Plant Science* 14 (2023): 1234555.
- [16] Eliazer, M., Jerrell Franklin C, and Shridhar Shrivastava. "Using Temporal Convolutional Networks (TCN) Deep Learning Model for Crop Recommendation." *Shridhar, Using Temporal Convolutional Networks (TCN) Deep Learning Model for Crop Recommendation* (May 11, 2024) (2024).
- [17] Gunasekaran, Hemalatha, Krishnamoorthi Ramalakshmi, Saswati Debnath, and Deepa Kanmani Swaminathan. "Physics-Aware Ensemble Learning for Superior Crop Recommendation in Smart Agriculture." *Sensors* 25, no. 19 (2025): 6243.
- [18] Prity, Farida Siddiqi, MD Mehadi Hasan, Shakhawat Hossain Saif, Md Maruf Hossain, Sazzad Hossain Bhuiyan, Md Ariful Islam, and Md Tousif Hasan Lavlu. "Enhancing agricultural productivity: a machine learning approach to crop recommendations." *Human-Centric Intelligent Systems* 4, no. 4 (2024): 497-510.
- [19] Mahale, Yashashree, Nida Khan, Kunal Kulkarni, Shivali Amit Wagle, Preksha Pareek, Ketan Kotecha, Tanupriya Choudhury, and Ashutosh Sharma. "Crop recommendation and forecasting system for Maharashtra using machine learning with LSTM: a novel expectation-maximization technique." *Discover Sustainability* 5, no. 1 (2024): 134.
- [20] Afzal, Hadeeqa, Madiha Amjad, Ali Raza, Kashif Munir, Santos Gracia Villar, Luis Alonso Dzul Lopez, and Imran Ashraf. "Incorporating soil information with machine learning for crop recommendation to improve agricultural output." *Scientific Reports* 15, no. 1 (2025): 8560.
- [21] Venkateswara, Stella Mary, and Jayashree Padmanaban. "Interpretable deep learning models for independent fertilizer and crop recommendation." *Scientific Reports* 15, no. 1 (2025): 41721.

- [22] Mohamed, Salma A., Olfat O. Abdel Maksoud, Abdelrahman Fathy, Ahmed S. Mohamed, Khaled Hosny, Hatem M. Keshk, and Sayed A. Mohamed. "A hybrid deep learning and rule-based model for smart weather forecasting and crop recommendation using satellite imagery." *Scientific Reports* 15, no. 1 (2025): 36102.
- [23] Ahmed, Rania A., Walid El-Shafai, Zeinab A. Ahmed, El-Sayed M. El-Rabaie, and Fathi E. Abd El-Samie. "High-precision crop recommendation system with stacking ensemble classifiers for optimizing agricultural productivity." *Scientific Reports* (2025).
- [24] Shastri, Sourabh, Sachin Kumar, Vibhakar Mansotra, and Rohit Salgotra. "Advancing crop recommendation system with supervised machine learning and explainable artificial intelligence." *Scientific Reports* 15, no. 1 (2025): 25498.
- [25] Nimmagadda, Satyanarayana. "Real-Time Crop Prediction and Recommendation Using Machine Learning for Enhanced Agricultural Productivity." (2025): 1-13.
- [26] Venkatanaresh, M., and I. Kullayamma. "Deep learning based concurrent excited gated recurrent unit for crop recommendation based on soil and climatic conditions." *Multimedia Tools and Applications* 83, no. 24 (2024): 64109-64138.
- [27] Gopi, P. S. S., and M. Karthikeyan. "Red fox optimization with ensemble recurrent neural network for crop recommendation and yield prediction model." *Multimedia Tools and Applications* 83, no. 5 (2024): 13159-13179.
- [28] Sravanthi, Gunaganti, and Nageswara Rao Moparthi. "An efficient IoT based crop disease prediction and crop recommendation for precision agriculture." *Cluster Computing* 27, no. 5 (2024): 5755-5782.
- [29] Anandamurugan, S. "Hybrid deep learning with a fusion model of weather and soil-based multi-crop recommendations for improving crop sustainability." *Theoretical and Applied Climatology* 156, no. 11 (2025): 1-25.
- [30] Shams, Mahmoud Y., Samah A. Gamel, and Fatma M. Talaat. "Enhancing crop recommendation systems with explainable artificial intelligence: a study on agricultural decision-making." *Neural Computing and Applications* 36, no. 11 (2024): 5695-5714.